



# Handbook for Theses and Dissertations

The Graduate School of the University of Utah  
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**Prepared by the Thesis Office**

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# Getting started with your thesis or dissertation

## What this handbook is for

The *Handbook for Theses and Dissertations* is your official guide for preparing and submitting a thesis or dissertation at the University of Utah. It outlines formatting requirements, submission procedures, and key university policies. You should use this handbook throughout your writing and submission process to ensure your manuscript meets all Graduate School requirements.

## Who you'll work with

Throughout the process, you'll work with several people:

- **Your supervisory committee:** Approves your topic, provides feedback on your research and writing, conducts your oral defense, and approves the final manuscript content.
- **Department chair or dean:** Confirms that departmental requirements have been met and approves the final reading.
- **The Thesis Office:** Reviews your manuscript formatting and issues final format approval before publication.

[Contact the Thesis Office](#) early with questions about formatting, style guide selection, copyright permissions, university regulations, or accessibility.

## Your responsibilities

You are responsible for managing your progress and ensuring your manuscript meets all university requirements.

- Follow University of Utah regulations for theses and dissertations
- Meet all deadlines for submission and graduation
- Use a department-approved citation style
- Ensure your manuscript follows the formatting standards in this handbook

## Process overview

Below is a high-level overview of the thesis or dissertation process. See the [Step-by-step instructions](#) section for detailed guidance.

1. **Choose your topic** with your committee
2. **Plan your timeline** for research, writing, and submission
3. **Write using your department's style guide** and this handbook for formatting
4. **Submit your work** for preliminary review (optional)
5. **Defend your thesis** or dissertation in a public oral exam
6. **Revise, gather approvals, and submit** your final manuscript

***Tip:** Schedule your defense and submission early. Strict deadlines apply for format approval and ProQuest upload.*



# Thesis and dissertation process

## Step-by-step instructions

This section outlines the steps from initial planning through final submission and graduation. Follow each step carefully and plan your timeline in advance.

### 1. Plan your timeline

Work with your committee chair to develop a schedule for research, writing, and revisions. Allow sufficient time for review, editing, and formatting corrections.

You are encouraged to submit your manuscript within two weeks of your defense. Graduation in a specific semester is not guaranteed if submission is delayed.

### 2. Consult with the thesis office (as needed)

Contact the Thesis Office early if you have questions about:

- Selecting a style guide
- Using illustrations or copyrighted materials
- Font or print specifications
- General formatting
- Accessibility requirements

You may request a preliminary review of one formatted chapter before your defense. This review helps identify formatting issues early.

- Only one chapter will be reviewed
- Submit as a PDF via email
- This review does not count as official submission

After your defense, manuscripts are accepted for format approval only.

### 3. Share your manuscript with your committee

Before your defense, submit the manuscript to:

- **Committee chair:** Must receive the full manuscript at least three weeks before the oral exam
- **Committee members:** Must receive the manuscript at least two weeks before the oral exam

**Note:** Some departments require more time, so confirm timelines with your chair.

#### 4. **Defend your work**

Complete a public oral defense, scheduled and announced by your committee.

- Schedule your defense early enough to make revisions and meet final deadlines
- Defense must be completed before your manuscript can be submitted for approval
- Approvals must be submitted no later than two weeks before the end of the semester (the last day of final examinations).
- You must be registered for at least one course from the time of formal admission through completion of all requirements for the degree, including the defense, unless granted an official leave of absence.
- Your oral defense must be publicly announced and open.

#### 5. **Revise your manuscript**

After the defense:

- Make all revisions required by your committee
- Obtain permission from your committee chair to submit your final version
- Submit the manuscript to your department chair or dean for approval
- Gather required uNIDs for digital signature forms (available in CIS under the Committee tab)

#### 6. **Collect digital approvals**

After submitting your manuscript in the university system:

- Digital signature forms are sent to your committee
- Required approvals include:
  - Supervisory Committee Approval Form (majority of committee)
  - Final Reading Approval Form (committee chair and department chair or dean)
  - Final Approval Form (Dean of the Graduate School, after format approval)

You will receive confirmation emails as signatures are completed. Follow up promptly on any missing approvals.

#### 7. **Submit for format approval**

Submit a PDF of your revised and approved manuscript to the Thesis Office.

- Submit at least two weeks before the end of the semester for same-term graduation
- Manuscripts are reviewed in the order received
- Formatting must meet handbook requirements for the submission to be considered

complete

Once approved, you will receive official Format Approval.

#### 8. **Complete corrections (if needed)**

If formatting issues are identified:

- Make required corrections promptly
- Schedule an in-person or virtual appointment if needed

Format Approval is issued once all corrections are approved.

#### 9. **Upload to ProQuest**

After receiving Format Approval:

- You'll receive instructions for submitting your PDF to ProQuest
- Upload your final file no later than the Wednesday before the last day of finals

Once the Thesis Office and ProQuest approve the final form, the Dean of the Graduate School signs it, and a thesis release is issued.

**Note:** *If your thesis release isn't issued by the semester deadline, you'll need to reapply for graduation for the next term.*

## **Electronic publication and archiving**

After approval, your manuscript is archived in:

- ProQuest (required)
- USpace (optional University of Utah repository)

USpace provides:

- Open access to your work
- Permanent URLs
- Indexing by search engines such as Google

You may grant USpace permission during the ProQuest submission process.

An abstract of your manuscript is also included in the University of Utah Abstracts of Theses and Dissertations, archived in the Marriott Library's Special Collections.

## **Publishing options and fees**

- ProQuest Traditional Publishing: Free
- ProQuest Copyright Registration (optional): \$55
- USpace Open Access: Free (with permission granted)

The Thesis Office does not collect fees. All payment is made through ProQuest.

## Statement of completion

If you need official proof that you've completed your degree requirements before your diploma arrives, you can request a Statement of Completion from the Office of the Registrar.

### **This document is available after:**

- Your thesis release has been issued by the Thesis Office
- Your graduate records file is complete
- All grades have been posted for your:
  - Application for Admission to Candidacy (master's students), or
  - Program of Study (doctoral students)

### **The Statement of Completion includes:**

- Your month of graduation
- The date you completed all degree requirements

**Note:** *Diplomas are mailed approximately three months after the semester ends. The Statement of Completion can be used in the meantime for job applications, licensing, or other official needs.*

# University requirements and policies

This section outlines institutional rules that apply to all theses and dissertations submitted at the University of Utah.

## Topic approval

Your supervisory committee must approve your thesis or dissertation topic before you begin writing. The topic must allow for eventual public access to the completed manuscript.

**Note:** See [Embargoes and delayed publication of theses and dissertations](#) for policies on restricted access.

## Contract research

If your research is funded by a sponsor or contract, it still must be made publicly available. The University may delay publication for up to six months with the sponsor's written approval.

### Excerpt from the University's standard research agreement:

... as a state institution of higher education, engages only in research that is compatible, consistent and beneficial to its academic role and mission and therefore significant results of research activities must be reasonably available for publication. The University agrees, however, for a period not to exceed six (6) months following completion of the project, that it will obtain sponsor approval prior to publication, which approval will not be unreasonably withheld by sponsor. The University agrees to keep confidential any sponsor proprietary information supplied to it by sponsor during the course of research performed by the University, and such information will not be included in any published material without prior approval by sponsor.

*(Office of the University of Utah Vice President for Research)*

## Oral defense requirements

Your committee will schedule a public oral exam where you must defend your thesis or dissertation. The defense must be publicized.

- The departmental director of graduate studies and the supervisory committee determine how the oral defense is publicized
- The supervisory committee grants manuscript approval only after a successful defense

## Publication requirements

Doctoral dissertations and master's theses must be available to other scholars and to the general public.

**Meet the publication requirement by doing one of the following:**

- Submit the entire dissertation or thesis to ProQuest where it will be made publicly accessible.
- Submit the abstract only to ProQuest if the whole dissertation or thesis has already been formally published and distributed (not vanity published).

**Note:** *In this case, you may also choose to publish the full, previously published version*

**Tip:** *Your manuscript must comply with university formatting requirements even if it includes previously published work.*

## **Submission of thesis or dissertation for publication**

After format approval, your final manuscript is submitted through ProQuest and archived in University of Utah library systems for public access.

## **Embargoes and delayed publication of theses and dissertations**

In some cases, the public release of a thesis or dissertation may be delayed to protect sensitive information. All embargo (delayed release) requests must be formally approved.

### **Reasons for requesting a delay**

- To protect the rights of patent applicants
- To prevent unjust economic exploitation
- To protect the privacy of research subjects
- To avoid copyright conflicts

### **Who must approve**

- Your committee chair (with knowledge of your department chair)
- The Dean of the Graduate School, in consultation with the Graduate Council

**Note:** *A delay should not exceed three years without additional approval.*

## **Embargo via ProQuest**

- You may request a two-year embargo of your thesis or dissertation when submitting it to ProQuest.
- To extend to a third year, contact the Thesis Office Manager after receiving Format Approval and uploading your manuscript

**Tip:** *For more guidance, see the [Resources](#) section of the Thesis Office website.*

## **Copyright and permissions**

U.S. copyright law is complex. It protects your work but also restricts your use of others'

content. It's important to understand both.

**Tip:** Visit [copyright.gov](http://copyright.gov) or [lib.utah.edu/research/copyright.php](http://lib.utah.edu/research/copyright.php) for detailed guidance.

### Your responsibilities regarding copyright

- Obtain written permission early, before you submit for review
- Keep proof of permission to submit to the Thesis Office
- Be prepared to replace copyrighted material if permission is denied

**Note:** *“Fair use” is a legal defense and does not guarantee the right to use copyrighted material without permission.*

### What copyright protects

You automatically own the copyright to your thesis or dissertation once it's written. You are also responsible for respecting the rights of others.

### You must obtain permission to use:

- Long quotations (more than 1.5 single-spaced pages)
- Reproduced materials (e.g., journal articles, figures, tables, survey instruments)
- Internet content (treated the same as printed material)

**Note:** *Even if you created the content yourself (such as a published article), you may still need permission if you transferred the rights during publication.*

*See Table 1.1 for general guidelines on determining whether a material is likely to be copyrighted based on age, authorship, and publication status.*

### Notice of copyright

You must include a copyright notice on the second page of your manuscript. This notice asserts your ownership of the work.

### Use the following format:

|                                                                                                               |
|---------------------------------------------------------------------------------------------------------------|
| <p>Copyright © <b>Your Name</b> (as it appears in CIS/the title page) <b>Year</b><br/>All Rights Reserved</p> |
|---------------------------------------------------------------------------------------------------------------|

See [Copyright page](#) for details on formatting and layout. This page is uncounted and unnumbered, but it must appear in every copy of your thesis or dissertation.

### Registration of copyright

Registration is optional, but it does offer additional legal protection.

### Why register?

- It places details of a copyright claim on public record.
- It qualifies you for statutory damages in cases of infringement

### **How to register**

- File online at [copyright.gov](https://copyright.gov), or
- Use ProQuest's optional registration service when submitting your manuscript

**Note:** *Paper forms may still be submitted to the U.S. Copyright Office, Library of Congress, Washington, DC 20559.*

### **Avoiding infringement**

You must obtain permission to use any material that falls outside fair use or copyright exemptions.

### **When permission is required**

You must get permission if you plan to use:

- A figure or table in its entirety
- A poem, musical composition, or lyrics
- A questionnaire, survey, or large excerpt of prose
- Any material from another source that exceeds what's allowed by fair use

### **Things to keep in mind**

Even if the material is short or permission is granted, you must:

- Cite the source properly in your manuscript
- Avoid any appearance of plagiarism

### **How to get permission**

#### **1. Identify the copyright holder**

This may be the publisher or the original author

#### **2. Request permission in writing**

Follow the publisher's formal process or use the form on the Thesis Office website:

[Permission to Quote Copyrighted Material](#)

#### **3. Submit proof of permission**

Email your documentation to:

- [thesis@gradschool.utah.edu](mailto:thesis@gradschool.utah.edu), or
- Your assigned manuscript editor



**Tip:** Always use the “credit line” provided by the copyright holder and place it where they specify.

## Use of journal articles in theses or dissertations

### Obtaining approval

If you plan to include articles written for publication in your thesis or dissertation, or use a journal’s formatting style, you must get approval from your supervisory committee. This includes:

- Reformatting your manuscript to match the style of a journal
- Including published, submitted, or accepted journal articles as chapters or sections

For details on how to include published or in-preparation articles, see [Appendix A: Using Published Articles as Dissertation Chapters](#).

### Use of previously published material

You may include previously published work in your thesis or dissertation, such as journal articles, essays, or other scholarly content, as long as specific guidelines are followed.

#### Requirements for inclusion

- Your supervisory committee must approve the inclusion
- The material must meet university standards for formatting and academic quality
- Published content must be treated as separate chapters or sections and integrated into your manuscript structure and formatting rules
- Pages from published articles must be renumbered to match your manuscript’s page sequence
- Content must adhere to university margin and layout rules, even if scanned or inserted as an image
- If you use more than one article, they must be thematically connected
- You must include an overall abstract summarizing the full work – even if individual chapters have their own abstracts

### Citation style

- Each published chapter may follow its original citation style
- Be consistent within each chapter, and clearly document which style guide is used

### Permission and attribution

- You must obtain permission to reuse copyrighted material
- A multiple author release must be submitted for any coauthored content

### Use of material accepted for publication

You may include articles that have been accepted for publication (but not yet published) in your thesis or dissertation, as long as they follow university formatting guidelines.

### Formatting requirements

- Manuscript pages must meet all university formatting rules (including margins, headings, spacing, table/figure placement)
- Running heads (displaying the article title and author in the page header) are not allowed
- Tables and figures must follow the placement rules in this handbook

### Structure and numbering

- Articles are treated as chapters or major sections
- Tables and figures should be numbered locally (for example, Table 3.1, Figure 3.2)

**Note:** *If your article was submitted or accepted but not yet published, it should resemble the rest of your manuscript, not a journal reprint.*

### Property rights

As the author, you retain the copyright to your thesis or dissertation. However, your rights may be affected by contracts or agreements related to your research.

You own the work, unless:

- You've signed a publication agreement transferring rights to a publisher
- You've entered into a research or funding agreement that limits how the work can be shared

**Example:** If you signed over copyright to a journal that published your article, you may need permission to include that article in your manuscript.

**Check for:**

- Publisher agreements
- Research contracts (especially those involving patents or proprietary data)
- Sponsored project terms

***Tip:** If you're unsure about your rights, contact your advisor, the Thesis Office, or the Office of General Counsel at the University of Utah.*

The length of copyright protection depends on when the material was created and whether it was published. Use the table below to help determine whether permission is required.

For more details, see the U.S. Copyright Office's full guide: [Circular 15A](#)

**How to use the table**

If you're using older material, such as a map, photo, or diagram, check:

- **When** it was created or published
- **Who** owns the rights (individual or corporate)
- **Whether** copyright has expired

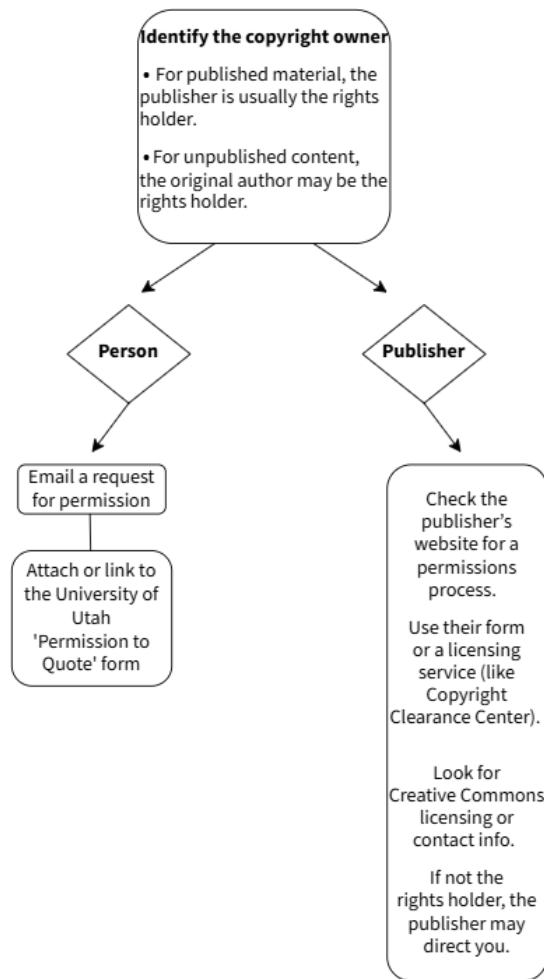
If you're not sure, treat the material as protected and seek permission.

**Table 1.1 Copyright guidelines**

Adapted from “Duration of Copyright Circular 15A” United States Copyright Office, U.S. Government Publishing Office, © copyright 2022

| Creation/Publication                                                                                          | General Rule of Copyright Duration                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Created in or after 1978 by a named author acting in an individual capacity, whether published or not.        | Life of the author, plus 70 years.                                                                                                                                                                       |
| Created in or after 1978 by an anonymous or pseudonymous author, a corporate author, or a work-made-for-hire. | The earlier of either 95 years from publication or 120 years from creation.                                                                                                                              |
| Created before 1978 but not published.                                                                        | The later of either 70 years after the author's death or December 31, 2002. The expiration date is extended through December 31, 2047, if the copyright owner publishes the work before the end of 2002. |
| Published after 1992 and before 1978 with copyright notice and renewed if required.                           | 95 years from the date of original publication.                                                                                                                                                          |
| Created and published before 1923.                                                                            | Copyright has expired.                                                                                                                                                                                   |

**Note:** The copyright circular can be viewed and downloaded for free in its entirety at: <https://www.copyright.gov/circs/circ15a.pdf>



**If permission is granted**

- Use the credit line as required by the rights holder.
- Submit all permission documentation to the Thesis Office.

**If the owner doesn't respond, you can**

- Adapt the material to create your own version.
- Replace it with different content.
- Remove it if necessary.

**\*\*Always document your attempt to contact the owner**

**Figure 1.1 How to request permission to reuse copyrighted material**

Use this chart to determine whether you need permission to reuse or adapt previously published content in your thesis or dissertation. Start by identifying the copyright holder (usually a publisher) and follow their official permissions process. If you receive permission, send documentation to the Thesis Office. If the copyright owner is unresponsive, you may need to adapt or remove the material. See the Thesis Office website for additional guidance and required forms.

## Content and authorship guidelines

### Content requirements

All theses and dissertations must demonstrate original research or creative work and meet academic standards set by your department, college, and supervisory committee.

### University expectations

Your content must:

- Present results of scientific or scholarly research or artistic creativity
- Show originality and independent investigation
- Make a contribution to knowledge or the creative arts
- Demonstrate mastery of the relevant literature
- Follow an acceptable academic style

*Source: University Regulations Library 6-203-III-F*

### Approvals required

- Your supervisory committee approves content, organization, and citation style
- The Thesis Office approves formatting and presentation

## Coauthorship

Some theses or dissertations may include work that was coauthored with others. This section outlines the conditions for including coauthored material.

### Material coauthored with nonstudent(s)

You may include work coauthored with nonstudents (e.g., faculty member or research director) if:

- The supervisory committee approves the inclusion
- Your role in the work meets university expectations for originality and independent investigation

**Note:** See *University Regulations Library 6-203-III-F* for full criteria.

## Coauthors as degree candidates

In rare cases, two or more students may submit a joint master's thesis. This is allowed only if:

- The supervisory committee gives prior approval
- Each student's contribution is clearly defined
- The thesis reflects the scope and depth expected of multiple authors
- The work represents independent, original contributions from each student

**Note:** *Doctoral students may not submit a coauthored dissertation.*

## Multiple author releases required

If you include any coauthored work in your thesis or dissertation, whether with students or nonstudents, you must submit a signed release form from each coauthor.

### Requirements

- A release is required even if you are the first author or the coauthor is your advisor or committee member
- The release confirms agreement to include the work in your manuscript

### How to submit

- Complete the electronic submission form on the Thesis Office website
- Submit the form before graduation

**Tip:** *You'll need accurate email addresses for each coauthor when completing the online form.*

## Approval requirements

Before your thesis or dissertation can be submitted for final release, all required approvals must be received.

### Required approvals

- **Supervisory committee members**  
Sign the Supervisory Committee Approval form (majority signature required)
- **Final reader (usually your committee chair) and department chair or dean**  
Sign the Final Reading Approval form to confirm content and formatting approval
- **Thesis Office manuscript editor**  
Issues format approval after all formatting corrections are complete

# Manuscript formatting and structure

This chapter explains the required structure and formatting for your thesis or dissertation. It includes details about each section, from the title page to appendices, and how to meet university-wide formatting requirements.

## Your manuscript must:

- Be written in clear, formal academic language
- Use a department-approved citation style
- Follow University of Utah formatting rules for layout, margins, fonts, and structure

You are responsible for meeting these formatting standards. Even if your committee approves the content and style, the Thesis Office may require changes before final approval.

## Documentation and writing style

You must use a department-approved style guide for citations and references. Journal-formatted chapters may follow the relevant journal style, but the overall manuscript must follow Graduate School formatting rules.

If the Thesis Office has not worked with your selected journal style before, you may be asked to provide a published article and/or the author's guide showing examples of figures, tables, and references.

If there is a conflict between the handbook and the style guide, the handbook takes priority for:

- Margins
- Line spacing
- Quotation placement
- Figure and table formatting

All writing must follow Standard American English unless your department approves an exception for creative works. Manuscripts with significant grammar, spelling, or punctuation errors may be returned unprocessed.

Proofread thoroughly to improve clarity and avoid delays. Tools such as Grammarly or other editing services may be used.

## Reference tools

Consult the following sources for grammar, usage, and spelling:

- *Webster's New Collegiate Dictionary*
- *Webster's Third New International Dictionary*
- *Fowler's Modern English Usage* (2004 edition)



## Parts of the thesis or dissertation

Your manuscript consists of three main sections:

1. **Front matter** – preliminary pages (title page, abstract, acknowledgments, etc.)
2. **Main text** – the body of your research, usually divided into chapters
3. **Back matter** – appendices, references, and your vita

Each part must follow university formatting guidelines, even if your content has been previously published or accepted elsewhere.

Not all sections are required. Your department or committee will determine which elements are appropriate for your program and research.

## Preliminary pages

The preliminary pages appear at the beginning of your manuscript. Some are required, while others are optional based on your committee's preferences or your degree type.

### All preliminary pages

- Use lowercase Roman numerals for page numbering (i, ii, iii...)
- Must follow the order and formatting rules listed below

### Required preliminary pages

1. **Title page** – Always page i (but no page number is shown)
2. **Copyright page** – Uncounted and unnumbered (appears after title page)
3. **Statement of approval** – Page ii
4. **Abstract** – Usually page iii

### Optional preliminary pages (include if approved by your committee)

- Dedication
- Epigraph
- Table of contents
- List of tables
- List of figures
- List of symbols, abbreviations, or definitions
- Acknowledgments
- Preface

**Tip:** Even though the title and copyright pages are not numbered, they still count when calculating page order. That means the abstract is usually page iii, even though it appears to be the first page

with a visible number.

### Understanding preliminary page formatting

Preliminary pages follow a unique set of formatting rules. Some are numbered, some are counted but unnumbered, and some are left out of the Table of Contents entirely. This can be confusing, especially when setting up page numbers in Word.

The table below shows which pages are required, counted, numbered, and included in the Table of Contents.

**Table 2.1 Preliminary page requirements and formatting**

| Page Type                                 | Required? | Counted? | Numbered? | Included in TOC?                    |
|-------------------------------------------|-----------|----------|-----------|-------------------------------------|
| Frontispiece                              | No        | Yes      | No        | No                                  |
| Title Page                                | Yes       | Yes      | No        | No                                  |
| Copyright Page                            | Yes       | Yes      | No        | No                                  |
| Statement of Approval                     | Yes       | Yes      | Yes       | No                                  |
| Abstract                                  | Yes       | Yes      | Yes       | First page no, additional pages yes |
| Dedication                                | Optional  | Yes      | No        | No                                  |
| Epigraph                                  | Optional  | Yes      | No        | No                                  |
| Table of Contents                         | Yes       | Yes      | Yes       | First page no, additional pages yes |
| List of Tables                            | If needed | Yes      | Yes       | First page no, additional pages yes |
| List of Figures                           | If needed | Yes      | Yes       | First page no, additional pages yes |
| List of Symbols/Abbreviations/Definitions | If needed | Yes      | Yes       | First page no, additional pages yes |
| Acknowledgments                           | Optional  | Yes      | Yes       | First page no, additional pages yes |
| Preface                                   | Optional  | Yes      | Yes       | First page no, additional pages yes |

**Note:** “Counted” means the page contributes to the total page count and affects the sequence of Roman numerals, even if the page number is not printed. Page numbers in the preliminary pages appear centered on the bottom of the page in lower-case Roman numerals. This differs from page numbers in the text, which appear on the top right of the page and use Arabic numerals.

## Title page

The title page is the first official page of your manuscript. It contains key information about your work and must follow strict formatting guidelines.

### Title page requirements

- This is **page i**, but the number is not displayed
- Use ALL CAPS for the title
- Center everything horizontally and vertically on the page
- Use consistent spacing between each section

Your title page must include, in this order:

1. **Title** – In ALL CAPS
2. **Your full name** – Two lines below the title (not in all caps)
3. **Submission statement**

A thesis or dissertation submitted to the faculty of The University of Utah  
in partial fulfillment of the requirements for the degree of

4. **Degree information**
  - Degree only (e.g., *Master of Arts*)
  - Or degree in program (e.g., *Doctor of Philosophy in Chemistry*)
5. **Department or program name**
6. **The University of Utah**
7. **Month and year of graduation** – Centered two lines below the university name

Use standard fonts (like Times New Roman 12 pt) and maintain university margins: 1.25” left/right, 1” top/bottom.

**Sample Title Page (Master's Thesis)**

MAXIMIZING THE EFFICIENCY OF A MASTER-PLANNED COMMUNITY

by

Alan T. Patterson

A thesis submitted to the faculty of The University of Utah  
in partial fulfillment of the requirements for the degree of

Master of Science

Department of Civil and Environmental Engineering  
The University of Utah  
December 2024

**Sample Title Page (Doctoral Dissertation with Different Program Name)**

A NEW SET OF ALGORITHMS FOR INCREASING MEMORY CAPACITY

by

Jennifer Lynn Stanford

A dissertation submitted to the faculty of The University of Utah

in partial fulfillment of the requirements for the degree of

Doctor of Philosophy in Computer Science

School of Computing

The University of Utah

August 2024

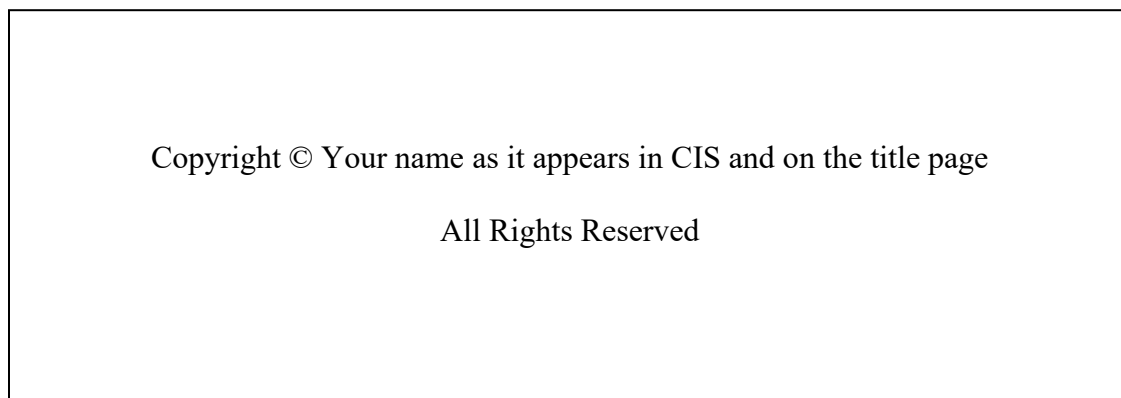
## Copyright page

The copyright page comes immediately after the title page. It affirms your legal ownership of the work.

### Formatting guidelines

- This page is uncounted and unnumbered
- Center the content both vertically and horizontally on the page
- Use the exact format below, adjusting only for your name and year
- Leave a double space between the two lines

### Example copyright page



**Tip:** To center vertically in Word, go to **Layout > Page Setup > Layout tab**, and under **Vertical alignment**, select **Center**. Make sure this applies only to the copyright page.

**Note:** This page must appear in every copy of your thesis or dissertation, even if it's not listed in the Table of Contents.

## Statement of Thesis or Dissertation Approval

This page confirms that your manuscript has been reviewed and approved by your supervisory committee. It must follow a strict layout and wording.

### Formatting guidelines

- This is **page ii** (and the first numbered page)
- Use standard margins and fonts (like Times New Roman, 12 pt)
- The content must be centered horizontally, but not vertically

**What to include**

1. The title of your thesis or dissertation (in ALL CAPS)
2. Your full name
3. A statement of approval using the following structure:

*Approved by*

*Supervisory Committee:*

*[Typed full legal names of all committee members]*

*[Typed full legal name of department chair or dean]*

*The University of Utah*

*Month, Year*

**Note:** Use the full legal names as listed in your CIS page under the Committee tab. Do not include titles (like Dr., PhD).

**Digital submission notes**

- You do not need handwritten signatures
- Signatures are collected electronically through the university system
- This page lists the names of your committee, department chair, and/or dean
- Use full legal names from your CIS Committee tab
- Do not list titles like Dr. or PhD

## Example Statement of Thesis Approval for master's students

### The University of Utah Graduate School

#### STATEMENT OF THESIS APPROVAL

The thesis of Kelly Harward  
has been approved by the following supervisory committee members:

Melanie Rae Thon, Chair 01/07/2014  
Date Approved

Lance Olsen, Member 01/07/2014  
Date Approved

Michael Mejia, Member 01/07/2014  
Date Approved

and by Vince Pecora, Chair/Dean of  
the Department/College/School of English

and by David B. Kieda, Dean of The Graduate School.



## Example Statement of Dissertation Approval for doctoral students

### The University of Utah Graduate School

#### STATEMENT OF DISSERTATION APPROVAL

The dissertation of Natalie J. Lecy  
has been approved by the following supervisory committee members:

|                                 |          |                                                      |
|---------------------------------|----------|------------------------------------------------------|
| <u>Joanna Bettmann Schaefer</u> | , Chair  | <u>date approved</u><br><small>Date Approved</small> |
| <u>Adam Hanley</u>              | , Member | <u>date approved</u><br><small>Date Approved</small> |
| <u>Tasha Keyes</u>              | , Member | <u>date approved</u><br><small>Date Approved</small> |
| <u>Mary Beth Vogel-Ferguson</u> | , Member | <u>date approved</u><br><small>Date Approved</small> |
| <u>Paul Rubin</u>               | , Member | <u>date approved</u><br><small>Date Approved</small> |

and by Philip Osteen, Chair/Dean of  
the Department/College/School of Social Work  
and by David B. Kieda, Dean of The Graduate School.

## Abstract

The abstract is a brief, stand-alone summary of your thesis or dissertation. It gives readers a concise overview of your research question, methods, findings, and conclusions.

### Formatting guidelines

- Appears after the statement of approval
- This is typically page iii
- The first page of the abstract is counted but not numbered
- Use the heading ABSTRACT (in all caps, centered 2 inches from the top)
- Double-space the body of the text
- Do not indent the first line

### If your abstract continues to a second page:

- Use lowercase Roman numerals (e.g., iv)
- Center the page number horizontally between the margins
- Place the number vertically between the bottom of the page and the top of the bottom margin

**Note:** *This formatting rule applies to any preliminary section that continues beyond one page.*

### Content requirements

- Master's students: 150–250 words
- Doctoral students: 350–500 words
- Must include:
  - Your research problem or objective
  - The methods used
  - Key findings
  - Main conclusions or contributions

**Tip:** *Write your abstract last, after your committee approves the final version of your manuscript.*

## Dedication or epigraph *(optional)*

These pages are optional, but many students include them to express appreciation or set the tone for their work.

### Dedication

A brief statement honoring someone special, often family, mentors, or friends.

- Center the text on the page (both horizontally and vertically)
- Use a simple, sincere tone
- Do not include jokes, graphics, or informal language
- Count the page in sequence, but do not show a page number

### Epigraph

A short quotation placed at the beginning of the manuscript or chapter to suggest a theme or mood.

- Optional and literary in nature
- Should be relevant to your work's subject, tone, or purpose
- Cite the author or source below the quote
- Follow same page numbering rule as dedication: counted, but not numbered

## Contents or Table of Contents

The Table of Contents (TOC) helps readers navigate your thesis or dissertation. It must accurately reflect your document's structure and follow university formatting standards.

### Formatting guidelines

- Label the page: CONTENTS (in all caps, centered, 2 inches from the top)
- Use period leaders to connect section titles and page numbers
- Align page numbers at the right margin
- Include only first-level headings and section titles (no sub-subheadings)

**Note:** *If the Table of Contents continues to a second page, repeat the heading **CONTENTS** at the top, centered 1 inch from the top margin (instead of 2 inches). Do not label it "continued" or use any alternate heading. Continue the entries without restarting page numbering.*

### What to include

List the following (if used):

- Abstract
- Acknowledgments
- List of Tables
- List of Figures
- List of Symbols, Abbreviations, or Definitions
- Preface
- Chapters (with full titles and starting page numbers)
- References
- Appendices
- Vita

Do **not** include:

- Title page
- Copyright page
- Statement of Approval
- Dedication
- Epigraph
- Table of Contents itself

***Tip:*** Use Word's heading styles to generate your Table of Contents automatically and ensure consistency.

**Example table of contents**

|                                    |     |
|------------------------------------|-----|
| CONTENTS                           |     |
| ABSTRACT .....                     | iii |
| ACKNOWLEDGMENTS .....              | iv  |
| LIST OF TABLES .....               | v   |
| CHAPTER                            |     |
| 1 INTRODUCTION .....               | 1   |
| 2 LITERATURE REVIEW .....          | 6   |
| 3 METHODS .....                    | 12  |
| 4 RESULTS AND DISCUSSION .....     | 21  |
| 5 CONCLUSION .....                 | 35  |
| REFERENCES .....                   | 38  |
| APPENDIX A: SURVEY QUESTIONS ..... | 42  |
| VITA .....                         | 58  |

### *List of tables*

Include a List of Tables if your thesis or dissertation contains two or more tables. This helps readers quickly locate key data or structured information.

#### **Formatting guidelines**

- Label the page: LIST OF TABLES (in all caps, centered, 2 inches from the top)
- Use period leaders to connect the table title and the page number
- Align page numbers at the right margin
- Double-space entries (unless your department allows single-spacing)
- Match the wording and capitalization of table titles exactly as they appear in the text

If the list continues to a second page, repeat the heading LIST OF TABLES at the top of the next page, centered 1 inch from the top. Do not add “continued” or modify the heading.

**Tip:** Use Word’s built-in styles and captions to generate this list automatically.

#### **Example list of tables**

| LIST OF TABLES                                      |    |
|-----------------------------------------------------|----|
| Table 1.1 Summary of participant demographics ..... | 5  |
| Table 2.1 Survey results by group .....             | 12 |
| Table 3.1 Regression analysis output .....          | 18 |

### *List of figures*

Include a List of Figures if your thesis or dissertation contains two or more figures (like charts, diagrams, illustrations). This list helps readers quickly locate visual elements in your work.

### Formatting guidelines

- Label the page: LIST OF FIGURES (in all caps, centered, 2 inches from the top)
- Use period leaders between figure titles and page numbers
- Align page numbers at the right margin
- Double-space entries unless otherwise directed by your department
- Match the wording and capitalization of figure titles exactly as they appear in the text

If the list continues to a second page, repeat the heading LIST OF FIGURES at the top of the next page, centered 1 inch from the top. Do not add “continued” or modify the heading.

**Tip:** Use Word’s caption tool to label your figures and auto-generate the list.

### Example list of figures

| LIST OF FIGURES                                             |    |
|-------------------------------------------------------------|----|
| Figure 1.1 Map of Study Area .....                          | 3  |
| Figure 2.1 Conceptual Framework Model .....                 | 9  |
| Figure 3.1 Flowchart of Participant Selection Process ..... | 14 |
| Figure 4.1 Graph of Results by Category .....               | 22 |
| Figure 5.1 Summary Diagram of Findings .....                | 31 |

### List of Symbols, abbreviations, or definitions *(optional)*

This list is optional, but recommended in technical, scientific, or mathematical writing. It provides a quick reference for recurring:

- Symbols
- Acronyms
- Abbreviations
- Units
- Specialized terminology

#### Formatting guidelines

- Use a descriptive title depending on the contents: LIST OF SYMBOLS, LIST OF ABBREVIATIONS, GLOSSARY, or a combined title (e.g., LIST OF SYMBOLS AND ABBREVIATIONS)
- Center the heading 2 inches from the top of the page in all capital letters
- Use lowercase Roman numerals for page numbering (counted and numbered)
- If the list continues onto a second page:
- Repeat the heading on the next page
- Center it 1 inch from the top

#### Layout

- Use consistent formatting for entries (e.g., tab leaders, columns, or aligned spacing)
- Alphabetize entries or group them logically
- Define all terms again in the text when they first appear—this list does not replace the requirement to define each term upon first usage.

### Acknowledgments *(optional)*

The acknowledgments section is optional. It's used to express gratitude to individuals or organizations who contributed to the research, writing, or support of your academic work.

#### Formatting

- The heading "ACKNOWLEDGMENTS" is centered and typed in all capital letters, 2 inches from the top of the page
- The acknowledgments text begins after a heading space



**Placement**

- If included, it follows the lists of tables, figures, abbreviations, and other preliminary material.
- If a preface is included, acknowledgments should be placed at the end of the preface, with no separate heading. In this case, do not list “Acknowledgments” separately in the table of contents.
- If there is no preface, acknowledgments appear as a standalone section and must be included in the table of contents.

**Preface** *(optional)*

A preface provides context about the origin, scope, or background of your research project. It may also include personal or academic motivations, acknowledgments of inspiration, or an explanation of how the work developed over time.

**When to include a preface**

- Your research has a unique origin story or interdisciplinary development
- You want to explain the circumstances or collaborations that shaped your project
- Your program or committee requires it

A preface is not required in most theses or dissertations, but it can be appropriate in certain fields or under specific guidance.

**Formatting guidelines**

- Label the page: PREFACE (in all caps, centered, 2 inches from the top)
- Use lowercase Roman numerals for the page number
- The first page is counted and numbered
- If the preface continues to a second page, repeat the heading on the next page, centered 1 inch from the top

## Formatting the main text

The main body of your thesis or dissertation presents your research in chapters or sections. This section explains how to format that content consistently, including headings, spacing, quotations, and page numbering. Even if your content varies by discipline, formatting must follow university guidelines.

### General formatting guidelines

- Use 12-point font, typically Times New Roman or a department-approved alternative (see table 2.0 below)
- Use double-spacing throughout the main text
- Indent the first line of each paragraph
- Use standard margins: 1.25" left/right, 1" top/bottom
- Avoid extra space between paragraphs

**Note:** Some word processing software adds extra space between paragraphs. Ensure spacing is consistent throughout the manuscript.

Table 2.2 Approved fonts and font sizes

| Font                 | Size             |
|----------------------|------------------|
| Arial                | 10 pt. or 11 pt. |
| Calibri              | 12 pt.           |
| Century              | 11 pt. or 12 pt. |
| Garamond             | 12 pt.           |
| Georgia              | 11 pt.           |
| Microsoft Sans Serif | 10 pt. or 11 pt. |
| Tahoma               | 10 pt.           |
| Times New Roman      | 12 pt.           |
| Trebuchet MS         | 10 pt. or 11 pt. |
| Verdana              | 10 pt.           |

### Page numbering in the main text

- Page numbering switches to Arabic numerals (1, 2, 3...) starting with Chapter 1
- Page 1 is the first page of the introduction or first chapter
- Page numbers should appear in the top right corner, 1 inch from the top and flush with the right margin
- Continue numbering consecutively through the end of the document

### Chapter and section structure

You may divide your manuscript into chapters or main sections, depending on your field and committee's preferences. Either format is acceptable if applied consistently.

- Chapters may use Roman or Arabic numerals, but not both
  - Example: CHAPTER I or CHAPTER 1
- Each chapter must have a title in all caps, centered
- If using section titles instead of numbered chapters, they may be:
  - Numbered (e.g., 1 INTRODUCTION)
  - Or unnumbered (e.g., INTRODUCTION, METHODS, RESULTS)

**Note:** All major titles (chapter or section) should be in ALL CAPS and match the Table of Contents exactly.

### Grammar and language standards

Your manuscript must follow Standard American English grammar and punctuation rules. Manuscripts with extensive grammatical errors will be returned unprocessed.

- Use clear, academic language appropriate to your discipline
- Exceptions may apply for creative works (with approval)

**Tip:** Using Grammarly or another grammar checker may help avoid delays in Thesis Release.

### Parts composed of related chapters (optional)

Some dissertations are divided into major sections called **Parts**, each containing a group of related chapters. This structure is optional and typically used when a project is split into multiple studies, themes, or phases.

### When to use parts

- Your dissertation includes distinct sections or case studies
- Your chapters are grouped under broader topics or methods

- You're combining independent but related articles or papers

### **Formatting guidelines**

- Label each part with a part title page (e.g., PART I: FOUNDATIONS)
- The title should be centered vertically and horizontally
- Start the first chapter of each part on the next page, with standard chapter formatting
- Do not restart page numbering (numbering continues through the entire document)

Part titles are not required in the Table of Contents, but if you include them, they must match the page and wording exactly.

## **Formatting and layout guidelines**

### **Quotations**

Quotations must follow standard academic formatting and citation rules. Be consistent with the style guide approved by your department (such as APA, MLA, Chicago, etc.).

#### **Short quotations**

- Quotations of fewer than three lines should appear in the main body of the text
- Enclose the quote in quotation marks
- Use punctuation and in-text citations according to your department's style guide

#### **Block quotations**

- Quotations of three lines or more must be formatted as block quotes
- Indent the entire block 0.5 inches from the left margin (same as the first line of a standard paragraph)
- Do not use quotation marks
- Single-space the quote; double-space before and after it
- Place the citation at the end, formatted per your citation style

## Headings and subheadings

Headings create structure and help readers navigate your manuscript. The university recognizes two types of headings: main headings and subheadings. This section explains how to format them, including capitalization, spacing, numbering, and accessibility.

Use 12-point Times New Roman or another standard serif font approved by your department for headings. All text, including headings and subheadings, must use the same font and size throughout the manuscript.

### *Main headings*

**Main headings are used for:**

- Chapters or major sections (e.g., CHAPTER 1, INTRODUCTION)
- Preliminary pages (e.g., ABSTRACT, ACKNOWLEDGMENTS, LIST OF TABLES)
- Final sections (e.g., REFERENCES, APPENDIX, VITA)

### **Formatting requirements**

- All main headings are in ALL CAPS
- Centered 2 inches from the top of the page
- Always begin on a new page
- Followed by a heading space (larger than double spacing)
- If the heading is more than two lines, double-space the lines
- If preceded by “CHAPTER” and a numeral, that line begins 2 inches from the top, followed by a heading space, then the chapter title
- If no subheading follows, text begins after a heading space
- If a subheading follows, begin the text after a double space
- Main headings longer than 5 inches should be split across lines in a visually balanced way.

### *Subheadings*

Subheadings organize the content within chapters or sections. There are two types:

- Freestanding subheadings: Stand alone on their own line
- Paragraph subheadings: Begin a paragraph of text on the same line

### **General rules**

- Use a consistent hierarchy – do not skip levels (e.g., don’t go from a first-level to a third-level subhead)
- Most manuscripts use one to three levels of subheadings
- Use the formatting scheme shown in your department guidelines or Figure 2.10 of the

manual

### **Capitalization**

- Do not use all caps for subheadings
- Use either headline-style capitalization or sentence-style
- Be consistent within each level of subheadings

### **Spacing and placement**

- Freestanding subheadings are preceded by a heading space and followed by a double space
- Paragraph subheadings are preceded by a double space only and are indented like a regular paragraph
- Subheadings of two or more lines are double spaced
- If a subheading appears at the bottom of a page, there must be room for:
  - The heading space
  - The subheading
  - At least two lines of text
    - If not, move the subheading to the top of the next page

### *Heading space*

A heading space is slightly larger than the standard double space. It is used:

- Below all main headings
- Above freestanding subheadings

Heading spaces are not used:

1. When a freestanding subheading begins a new page
2. Above a paragraph subheading
3. Between consecutive subheadings (use a double space only)

Regardless of your word processing software, heading spaces must be consistent throughout the manuscript.

### *Numbering subheadings (optional)*

Some departments use a numbering scheme for subheadings, especially in technical or scientific fields.

- Chapters are numbered with Arabic numerals
- First-level subheadings: 1.1
- Second-level: 1.1.1

- Third-level: 1.1.1.1, and so on
- If this system is used, all subheadings must be numbered consistently, including paragraph subheadings

### *Accessibility*

All headings and subheadings must be formatted using built-in styles or accessibility tags so that screen readers can recognize the document structure.

- If you're using the Thesis Office's official Word template, heading styles are already built in
- If you're using another program (e.g., LaTeX, InDesign), ensure your headings are tagged properly using that software's accessibility tools

Proper heading structure improves navigation for all users and is required for accessibility.

**Example main text**

CHAPTER 3  
METHODOLOGY

Main heading (centered, ALL CAPS, 2" from top)

[Heading space]

This chapter outlines the methodology used in the study...

[Double space]

**Participants** ← First-level subheading (freestanding)

[Double space]

This section provides an overview of the participants and relevant demographic details...

[Double space]

*Recruitment strategy* ← Second-level subheading (freestanding or paragraph)

Participants were recruited through university-wide email announcements...



## Print quality and formatting standards

All text must be clear and legible in the final PDF:

- Fonts must be professional and readable (e.g., Times New Roman 12 pt)
- Table 4.1 lists other acceptable fonts
- Font must be consistent throughout, including in page numbers
- Exceptions: programming code, content in figures or tables, or published materials in appendices

Justified right margins are permitted only if:

- Spacing is not excessive
- Hyphenation is used properly
- All text blocks (including captions and footnotes) are fully justified

No software guarantees correct formatting. You are responsible for ensuring compliance.

**Note:** For help with formatting in software programs, visit the Knowledge Commons in the Marriott Library.

## Accessibility and PDF formatting

You must ensure your final PDF is accessible to screen readers and meets digital accessibility standards.

- Use built-in heading styles, tagged captions, and alt text where applicable
- After conversion, test the PDF to confirm accessibility was preserved
- Improperly tagged PDFs may delay approval

## Footnotes

Footnotes may be used for commentary, citations, or explanatory information. They must follow university guidelines for formatting, margins, and spacing. Whether sources are cited in footnotes or through in-text references depends on the requirements of your discipline and department.

### Placement and formatting

- Footnotes must appear at the bottom of the same page as the text they reference
- A solid 20-space line separates the main text from the footnote area
  - This line must not extend into the left margin

- The first line of each note begins double-spaced below the solid line
- The note number is raised (superscript) and is not followed by a period or space

**Font, size, and margins**

- Footnotes must be in the same font style as the manuscript text
- A smaller font size is permitted if applied consistently
- The font size must be at least 2 millimeters high
- Footnotes must adhere to the same margin requirements as the rest of the manuscript

**Spacing**

- Footnotes must be single-spaced within entries
- Use a double space between entries
- If print quality is clear, footnote entries may use single spacing throughout, but spacing must remain consistent throughout the chapter

**Multiple footnotes on one page**

- When more than one footnote appears on the same page, each note must begin on that same page
- If a note is too long, the remaining content may continue in the footnote area of the next page
- Do not split a note across the bottom of one page and the top of the next

**Example footnote**

---

<sup>1</sup> Footnote sample. Note the font style is the same as the rest of the text. The size of the font may be the same size as the text or smaller, so long as it is consistent for all the footnotes.

<sup>2</sup> Footnote sample. Style guide used determines whether the footnote is inside or outside punctuation such as periods, commas, etc.

## Endnotes

- If your style uses endnotes, they must appear at the end of each chapter or main section, not at the end of the manuscript
- Each endnote section begins on a new page with a first-level subheading titled “Notes” or “Endnotes”
- Endnotes follow the same spacing rules as footnotes
- The word “Notes” or “Endnotes” must be listed in the table of contents as a first-level subheading

## Tables, figures, and equations

### Tables and figures in the text

Tables and figures help illustrate your research findings. They must be clearly labeled, numbered, and placed according to university formatting standards. This section outlines the formatting, placement, accessibility, and numbering rules for all tables and figures.

- A table is a compilation of data arranged in columns or rows.
- A figure is a visual or graphic presentation such as a photograph, map, diagram, plate, chart, or schematic.

### Each table and figure must:

- Be referred to by number in the text, in numerical order upon first mention (e.g., Table 1.1, Table 1.2, Figure 2.1, etc.)
- Be placed as close as possible to its first reference
- Be referenced by number only (not “the figure below” or “the table on the next page”)
- Be mentioned at least once, and appear only once in the manuscript (you may refer back to previously used items but may not duplicate them)

Refer to Appendix D for additional placement options and examples.

### Tables

#### Numbering

Tables may be numbered either consecutively throughout the manuscript (e.g., Table 1, 2, 3...) or by chapter using numbering by location (e.g., Table 2.1, 2.2).

If numbering by location is used, each chapter or major section must be numbered with Arabic numerals. In the appendices, tables are numbered A.1, A.2, B.1, etc. Numbering by location may not exceed one decimal place.

Tables may not be numbered by subheading.

## **Titles and formatting**

Table titles are placed above the table, in the same font and size as the rest of the text (12 pt Times New Roman). Titles may be single- or double-spaced but must be consistent throughout. A double space separates the title from the table.

Use solid horizontal lines below the title, after the column headings, and at the bottom of the table. Avoid vertical lines and thick-lined boxes unless required. Table text must be sharp and legible. Font size for the data may be reduced if necessary, but must remain readable.

If a table continues onto a second page:

- The table number and the word “continued” appear at the top of the continued page
- Column headings must be repeated

## **Placement**

Tables should be placed as close as possible to their first reference in the text. A common method is to group all tables at the end of the chapter or section under a first-level subheading titled “Tables.” If used, this grouping must be consistent across chapters.

Multiple tables on one page must be separated by more than double spacing.

Tables must fit within thesis margins (1.25" left/right, 1" top/bottom). Large tables may be reduced to fit, provided all text remains sharp and legible. When necessary, landscape (broadside) orientation may be used. If a table title is too long to fit with the table, use a part-title page.

## **Accessibility**

- Include alt text for each table
- Do not rely on color alone to convey meaning; use labels or patterns
- Use a minimum font size of 10 pt for table contents
- Table titles must be tagged or styled for screen readers
- If using software other than the official Word template, accessibility tools must be used

[See Appendix B](#) for further details about alt text and color use.

## **Oversize tables**

Oversize tables that cannot be reduced to fit standard pages must be submitted as:

- Two hard copies (one to the Thesis Office for archiving, one to the department)
- A digital version, if legible

## **Part-title pages**

When a table title is too long to fit with the table, place it on a separate part-title page:

- The title begins at or slightly above mid-page
- It is single-spaced (or double-spaced if superscripts/subscripts are used)

- Page numbers appear in the upper right corner
- The part-title page is listed in the List of Tables and included in the page count

## Figures

### *Numbering*

Figures follow the same numbering rules as tables: consecutive throughout or local by chapter. If local numbering is used, chapters must be numbered with Arabic numerals, and appendix figures use A.1, A.2, etc. Figures may not be numbered by subheading, and may not exceed one decimal place.

### *Captions and formatting*

Figure captions appear below the figure. The first sentence of the caption serves as the label in the List of Figures (if used). If the figure has multiple parts (a, b, c...), the caption must start with a general sentence, followed by sentences describing each part.

#### **Captions must:**

- Be in the same font and size as the body text (12 pt Times New Roman)
- Be single-spaced unless they include super- or subscript numbers
- Follow a consistent format throughout the manuscript
- Not be placed in a separate list above a group of figures

Figures must not be enclosed in thick-lined boxes. All print must be crisp and legible. Use color only when necessary, and always include high-contrast labels or patterns.

### *Placement*

Figures may be placed near their first mention or grouped at the end of each chapter under a first-level subheading titled “Figures.” This grouping must be consistent across chapters.

Figures should appear balanced on the page and centered horizontally within thesis margins. Multiple figures on one page must be separated by more than double spacing.

If a figure continues onto a second page:

- Place the figure number and the word “continued” at the bottom
- Use consistent formatting

### *Accessibility*

- Provide alt text for each figure
- Avoid using color alone for meaning
- Use the built-in caption styles or tags in Word or your chosen program
- Ensure minimum text size of 10 pt within figures
- Refer to [Appendix B](#) for guidelines on accessible figure design and descriptions

### *Landscape (broadside) placement*

Full-page figures may be rotated to landscape orientation. If the caption does not fit on the same landscape page, place it on a portrait-oriented part-title page preceding the figure.

### *Part-title pages*

If a figure caption is too long to fit with the figure, place it on a part-title page:

- Begin the caption at or slightly above mid-page
- Use single spacing (or double if super/subscripts are present)
- Include the page number in the upper right
- The part-title page should be included in the List of Figures and in the page count

### *Oversize figures*

Figures too large for standard pages must be submitted in hard copy:

- One copy for the Thesis Office (Special Collections)
- One copy for your department

Ensure digital versions remain legible.

### *Page numbering*

All pages containing figures must have a page number in the upper right corner, consistent with the rest of the manuscript.

## **Equations**

Equations must be formatted to meet university standards for placement, spacing, and numbering. Displayed equations should appear within thesis margins and must be clearly presented.

### **Formatting rules**

- Displayed equations must be centered between the margins
- Leave double spacing above and below each displayed equation
- Number only those equations that are referred to in the text
- Place equation numbers flush right, enclosed in parentheses
- Use local numbering by chapter (e.g., (1.1), (1.2), (2.1), etc.)
- Use Arabic numerals only. Do not use Roman numerals or outline-style numbering
- Refer to equations by number, not by placement (e.g., avoid “the equation above”)

## References or selected bibliography

All sources must be cited clearly and consistently using a department-approved style guide. The citation method you use depends on your discipline.

### Citation styles

Different academic disciplines use different citation styles. Check with your department for the required style. Common citation styles include:

- APA (American Psychological Association)
- MLA (Modern Language Association)
- Chicago or Turabian
- IEEE (Institute of Electrical and Electronics Engineers)
- Other field- or journal-specific styles

Only one citation style may be used in the manuscript. Do not mix styles unless explicitly permitted by your department.

### Citation methods

Citation methods include parenthetical references, footnotes or endnotes, and numbered systems.

- **Parenthetical references**
  - The author's name and publication date (or page number) appear in parentheses directly in the text
  - Common in APA and MLA styles
- **Footnotes or endnotes**
  - A superscript number in the text refers to a note at the bottom of the page or the end of the chapter
  - Common in Chicago style and some humanities disciplines
- **Numbered citation systems**
  - Source numbers may be presented in superscript, brackets, or parentheses depending on the style guide
  - Superscript numbers are placed outside punctuation marks
  - Bracketed or parenthetical numbers may appear inside or outside punctuation, depending on the style

- When citing multiple sources together, separate numbers with commas (e.g.,  
<sup>6, 7, 10-12</sup>)

## Reference list formatting

Each in-text citation must correspond to a complete entry in the reference list, works cited, or bibliography, depending on the citation style.

- Begin the reference list on a new page
- Use the appropriate title (e.g., REFERENCES, BIBLIOGRAPHY, WORKS CITED), centered 2 inches from the top
- Format each entry according to the chosen style guide (spacing, punctuation, ordering, indentation)
- If using a numbered citation system, the reference list must be in numerical order, not alphabetical
- In manuscripts composed of journal articles as chapters, a reference list may appear at the end of each chapter

## Placement and order

- The references or bibliography appear at the end of the manuscript
- Items are listed in numerical or alphabetical order, depending on the citation style used
- In scientific papers, only material actually cited in the manuscript is listed
- In humanities disciplines, background sources not directly cited may be included in a Selected Bibliography
- If the manuscript is a collection of journal articles as chapters, each chapter has its own reference section, formatted with the heading “REFERENCES” as a first-level subheading

## Heading and formatting

- The section is typically titled REFERENCES
  - Other acceptable headings include LITERATURE CITED or SELECTED BIBLIOGRAPHY, depending on the style used
- The heading must appear in all capital letters, centered, 2 inches from the top of the page
- A heading space follows before the first entry
- Entries must be:
  - Single spaced within each entry



- Double spaced between entries

### Part-title page option

If a part-title page is used before the references section (such as in article-style dissertations):

- The part-title page includes only the section title (e.g., “REFERENCES”), centered either 2 inches from the top or mid-page
- The first reference then begins on the next page, 1 inch from the top, and no main heading is repeated on that page
- The part-title page is counted in pagination but does not display a page number
- It is considered the first page of the section for Table of Contents purposes

### Table of contents

- The references section is the last item listed in the Table of Contents, unless a vita is included
- If a vita is included, it follows the references and becomes the final item listed

### Special note on et al.

- *Et al.* is short for *et alii* (Latin for “and others”)
- *Et* is not an abbreviation and should not be followed by a period
- *Al.* is an abbreviation and should be followed by a period
- Some style guides require *et al.* to be italicized; consult your selected style guide for specific requirements.

### Web citations

Web citations must include the author, title of publication, and date of publication, formatted according to the chosen style guide.

Include the full URL in the following format:

- Protocol or access-mode identifier (e.g., http, https), followed by a colon and two forward slashes
- Host name (always in lowercase), e.g., <http://www.ieee.org/>
- Pathway to the document, including file names, with exact upper/lower case and punctuation

**Important:** URLs may only be broken after a slash (/ or //) or after a period (.).

*Do not insert a hyphen, and avoid breaking a URL at an existing hyphen.*

## Appendices

Appendices present material that supports but is not essential to the main text. Appendices may include figures, tables, raw data, source material, computer programs, questionnaires, letters, or other types of supplemental information.

Appendices must appear after the references or bibliography. Each appendix must be referenced in the text and listed in the table of contents.

### Numbering and labeling

- If there is only one appendix, label it APPENDIX (in all caps)
- If there is more than one appendix, label them in the order they appear: APPENDIX A, APPENDIX B, APPENDIX C, etc.
- Titles should follow two lines below the label, also in all caps

### Example appendix label

|                                 |
|---------------------------------|
| APPENDIX A<br>SURVEY INSTRUMENT |
|---------------------------------|

### Page numbering

- Appendix pages are numbered consecutively with the rest of the manuscript
- Do not restart page numbers or use separate numbering for appendices
- Page numbers appear in the top right corner, consistent with the rest of the document

### Formatting and spacing

- Appendices must use the same font, margins, and character size (minimum 2 mm) as the body text
- Material may be single or double-spaced depending on its content
- If appendices contain subheadings, they must follow the same formatting and hierarchy used in the main text
- Subheadings within appendices are not listed in the table of contents

### **Tables and figures in appendices**

Tables and figures in appendices may be numbered in one of two ways:

- Continuously from the main text (e.g., Table 21, Figure 17)
- Locally, using the appendix letter as a prefix (e.g., Table A.1, Figure B.2)

Tables and figures must follow the same placement, caption, and formatting rules used in the main text.

### **Part-title pages for appendices**

A separate part-title page is often used to introduce each appendix when the content is diverse, complex, or reprinted. If one appendix uses a part-title page, then all appendices must use one for consistency.

- The appendix label (e.g., “APPENDIX A”) is centered and appears either two inches from the top or at mid-page
- The placement must be consistent across all appendices
- Part-title pages are counted in pagination but do not display a page number
- For table of contents purposes, the part-title page is treated as the first page of the appendix

### **Alternative formats and supplementary files**

Some materials, such as oversized content, scanned documents, or multimedia, may be excluded from the main manuscript and submitted separately to ProQuest.

- In these cases, include a part-title page in the manuscript that lists the file name and title of the omitted material
- The part-title page is paginated and listed in the table of contents

## **DVD submission**

If a department requires submission of a DVD (e.g., for multimedia content), submit two copies:

- One to the Thesis Office (for library archival purposes)
- One to your department

DVDs cannot be submitted electronically due to copyright restrictions. Check with your department for additional requirements.

## Vita

A vita may be included at the end of your manuscript, but it is not required. If used, the vita must follow the references or selected bibliography and appear as the final item in the manuscript. It must also be listed as a main heading in the table of contents.

The vita should follow a standard short curriculum vitae format, summarizing your professional academic background. It may include:

- Your full name
- Colleges and universities attended
- Degrees and certificates earned
- Professional organizations or affiliations
- Academic or professional positions held
- Relevant publications

Do not include personal information such as your birthdate, home address, or social security number.

Use a simple, professional tone. The vita is not a résumé or cover letter but a formal academic record.

**Note:** *The title of this section may be either VITA or CURRICULUM VITA (both are acceptable). However, VITAE is not correct.*

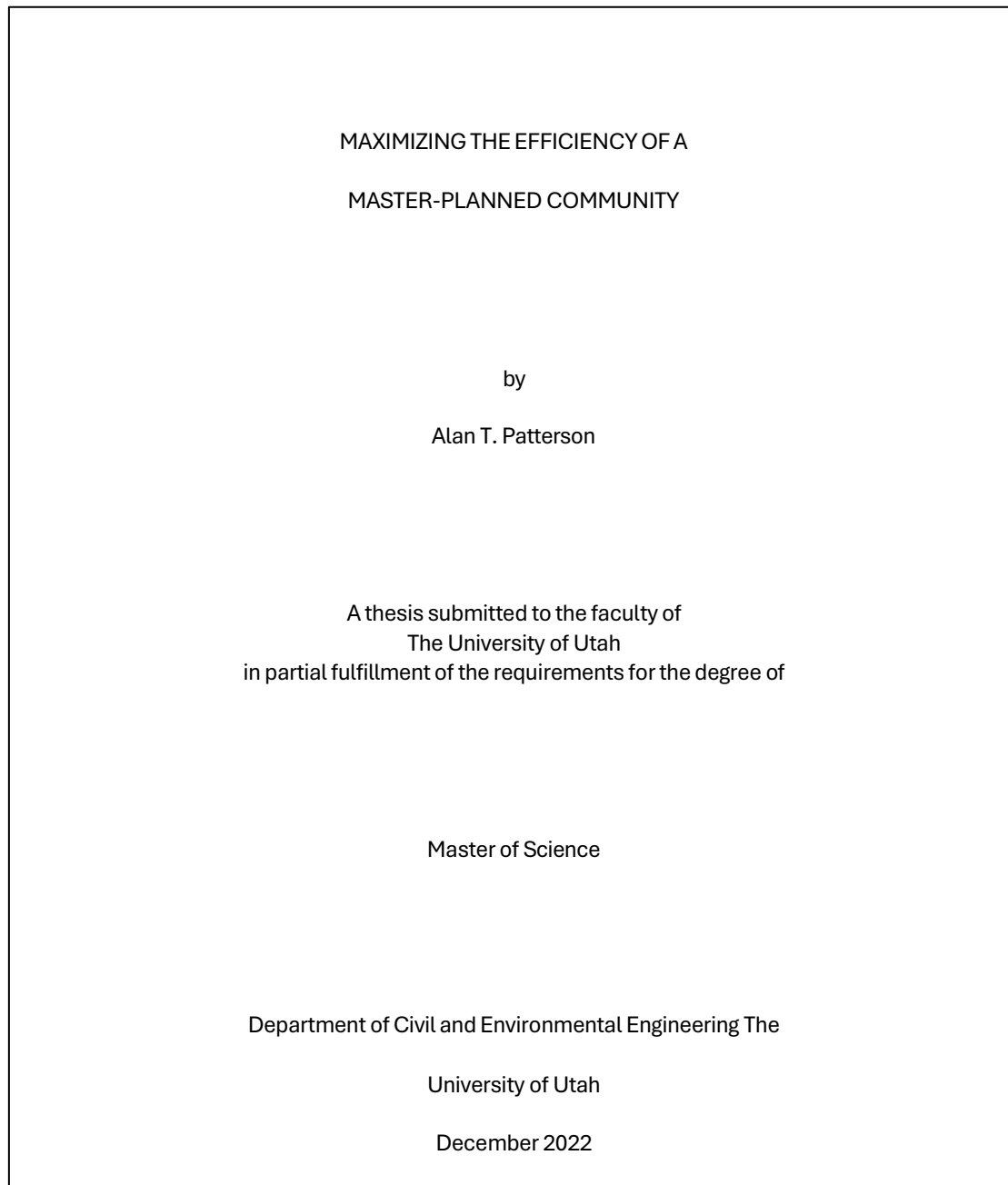


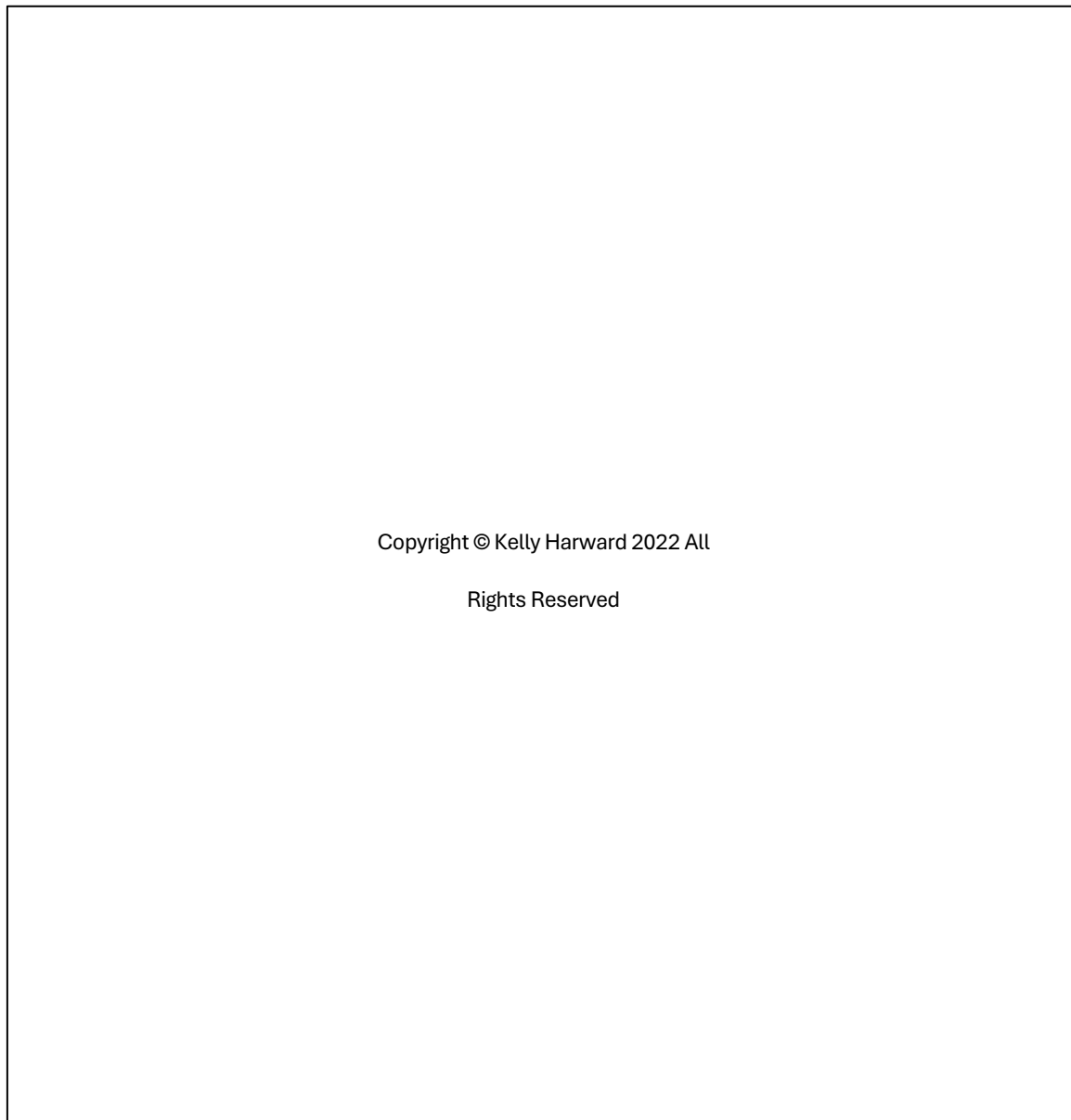
Figure 2.1 Sample title page of a master's thesis with degree and department the same.

If you're submitting a dissertation, simply change the word "thesis" to "dissertation." Note the spacing of elements on this sample title page and that the entire block of text is centered on the page.



Figure 2.2 Sample title page of a doctoral dissertation with a different degree and department.

If you're submitting a thesis, simply change the word "dissertation" to "thesis." Note the spacing of elements on this sample title page.



**Figure 2.3 Sample of a copyright page.**

The student's name should match what is listed in CIS, the title page of the thesis or dissertation, and the Statement of Approval. Note the spacing of elements on this sample copyright page.

**The University of Utah Graduate School**

STATEMENT OF THESIS APPROVAL

The thesis of Kelly Harward  
has been approved by the following supervisory committee members:

|                         |          |                                                   |
|-------------------------|----------|---------------------------------------------------|
| <u>Melanie Rae Thon</u> | , Chair  | <u>01/07/2014</u><br><small>Date Approved</small> |
| <u>Lance Olsen</u>      | , Member | <u>01/07/2014</u><br><small>Date Approved</small> |
| <u>Michael Mejia</u>    | , Member | <u>01/07/2014</u><br><small>Date Approved</small> |

and by Vince Pecora, Chair/Dean of  
the Department/College/School of English

and by David B. Kieda, Dean of The Graduate School.

**Figure 2.4 Sample page of the Statement of Thesis Approval for master's students.** Your name should match what is listed in CIS, the title page of the thesis, and the copyright notice. Note that the names are all typed (there are no signatures), and committee members' names do not include Dr., PhD, Professor, or any other such title.



**The University of Utah Graduate School**

**STATEMENT OF DISSERTATION APPROVAL**

The dissertation of Natalie J. Lecy  
has been approved by the following supervisory committee members:

Joanna Bettmann Schaefer , Chair date approved  
Date Approved

Adam Hanley , Member date approved  
Date Approved

Tasha Keyes , Member date approved  
Date Approved

Mary Beth Vogel-Ferguson , Member date approved  
Date Approved

Paul Rubin , Member date approved  
Date Approved

and by Philip Osteen , Chair/Dean of  
the Department/College/School of Social Work

and by David B. Kieda, Dean of The Graduate School.

**Figure 2.5 Sample page of the Statement of Dissertation Approval for doctoral students.**

Your name should match what is listed in CIS, the title page of the dissertation, the Supervisory Committee and Final Reading approval forms. Note the names are all typed (there are no signatures) and committee members' names do not include Dr., PhD, Professor, or any other such title.

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Figure 2.6 Sample table of contents using chapters and a local numbering scheme for subheads.

Note the spacing and that the word chapter is used only once. The margins have been cropped from this image. Adapted from Phelps, R. (2009). A two-step model of colon adenoma initiation and progressions that requires CtBP1 and KRAS following loss of APC, Doctoral Dissertation, the University of Utah, Salt Lake City.

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Figure 2.7 Sample table of contents using main sections rather than chapters. Note the spacing.

The margins have been cropped from this image. Adapted from Wessel, J. (2009). *The Muslim Brotherhood in Egypt and Jordan: A history of modern Islamic fundamentalism*, Master's Thesis, the University of Utah, Salt Lake City.

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Figure 2.8 Sample table of contents using main sections with Roman numerals.

Note the spacing. The margins have been cropped from this image.

species, is strongly supported as sister to *S. heterodoxum* var. *neomexicanum* in the ITS only (85% BS, 0.98 PP), *axay* only (87% BS, 1.0 PP) and combined analyses (89% BS, 1.0 PP), and, along with *S. citrullifolium* var. *knoblickii*, comprises a monophyletic group in the *axay* only (100% BS, 1.0 PP) and combined analyses (65% BS, 1.0 PP), here termed the Glandulosa clade.

**SPECIES- AND INTRASPECIFIC-LEVEL MONOPHYLY**—Species-level monophyly was examined in a number of taxa with multiple accessions sequenced in the phylogeny. In the cases of *S. nystrosum*, *S. grayi*, *S. ohlsonii*, *S. daviscowii*, *S. tribulosum*, and *S. leuipipes*, all accessions of the same species formed monophyletic groups with strong support in the combined trees. Furthermore, the multiple accessions sequenced of *S. citrullifolium* var. *citrullifolium* and *S. heterodoxum* var. *schlegelii* each emerged as monophyletic in all combined analyses, but *S. citrullifolium* var. *schlegelii* is paraphyletic in the combined MP strict consensus tree. However, *S. citrullifolium*, *S. heterodoxum*, and *S. grayi* var. *grandiflorum* were not supported as monophyletic, as multiple accessions of these taxa did not group together in the combined analyses.

**CONSTRAINT ANALYSES**—Constraining all of Whalen's series to be monophyletic resulted in trees significantly different than the most parsimonious tree from the combined dataset (Templeton's test  $p = 0.001$ ). When constraining *ser. Androceras* and *Volucrofolium* individually, the trees were also significantly different than the most parsimonious tree from the combined dataset ( $p = 0.0455$  and  $0.0477$ , respectively). Trees constraining all of the yellow-flowered taxa (i.e. species of *ser. Androceras* minus *S. tribulosum*) to monophyly were not significantly different than nonconstrained trees (Templeton's test  $p = 0.6698$ ).

#### DISCUSSION

**Sectional Relationships and Monophyly of Section Androceras**—Despite the various hypotheses regarding the sister group to sect. *Androceras*, our data support previous molecular studies in finding sect. *Crinium* as sister to sect. *Androceras* (Levin et al. 2006, Weese and Bohn 2007). These groups are morphologically distinctive and this relationship merits further study. *Solanum susymbriifolium* is sister to a clade composed of sect. *Androceras* and sect. *Crinium* despite the fact that *S. susymbriifolium* and sect. *Androceras* share highly divided leaves and strongly accrescent calyxes, characters not found in sect. *Crinium*. Lester et al. (1999) also found the seeds of sects. *Androceras* and *Cryptocarpum*, to which *S. susymbriifolium* belongs, to be remarkably similar. We were not able to sample other members of sect. *Cryptocarpum* but further sampling might possibly place this group sister to sect. *Androceras*.

All three data sets strongly support the monophyly of sect. *Androceras* as circumscribed by Whalen (1979a, 1984). This molecular evidence, combined with unique morphological traits found in the leaves, flowers, and fruits, its distinctive flavonoid chemistry, and geographical distribution, leave little doubt that *Solanum* sect. *Androceras* is a monophyletic group.

**Character Evolution and Monophyly of Whalen's Series in sect. Androceras**—Whalen's (1979a) three series within sect. *Androceras* were distinguished by trichome, flower, and seed morphology as well as flavonoid chemistry and geographical distribution (Table 1). Whalen (1979a) circumscribed these

series as natural phyletic groups; however, they were not defined in strict monophyletic terms (see paraphyly of *ser. Androceras* and *Volucrofolium* in Fig. 15 in Whalen 1979a). It is clear in examining his matrix of morphological characters (Table 6 and Fig. 15 in Whalen 1979a), that many are homoplasious or autapomorphic. Additionally, the assessment of ancestral and derived characters as well as coding of characters are based on the author's interpretations (see secondarily lost characters in Table 5 in Whalen 1979a) and could be differently interpreted by other taxonomists. Given this and the fact that our combined molecular dataset contains 340 parsimony informative characters, it is not surprising that the addition of the 14 characters from Whalen's (1979a) dataset does not change the topology or resolution of the phylogeny (results not shown). The few synapomorphic characters in Whalen's (1979a) character matrix show support for *ser. Pacificum*, the only one of the three series that emerges as a monophyletic group in our molecular trees. Characters unique to this series include white, deeply stellate corollas, radially wrinkled seeds, and a geographical center of distribution on the Pacific slope of the Sierra Madre Occidental on the west coast of Mexico. Apparently these characters arose once in the Pacificum clade, although confirmation of this awaits sampling of the third member of *ser. Pacificum*, *S. lowianum*.

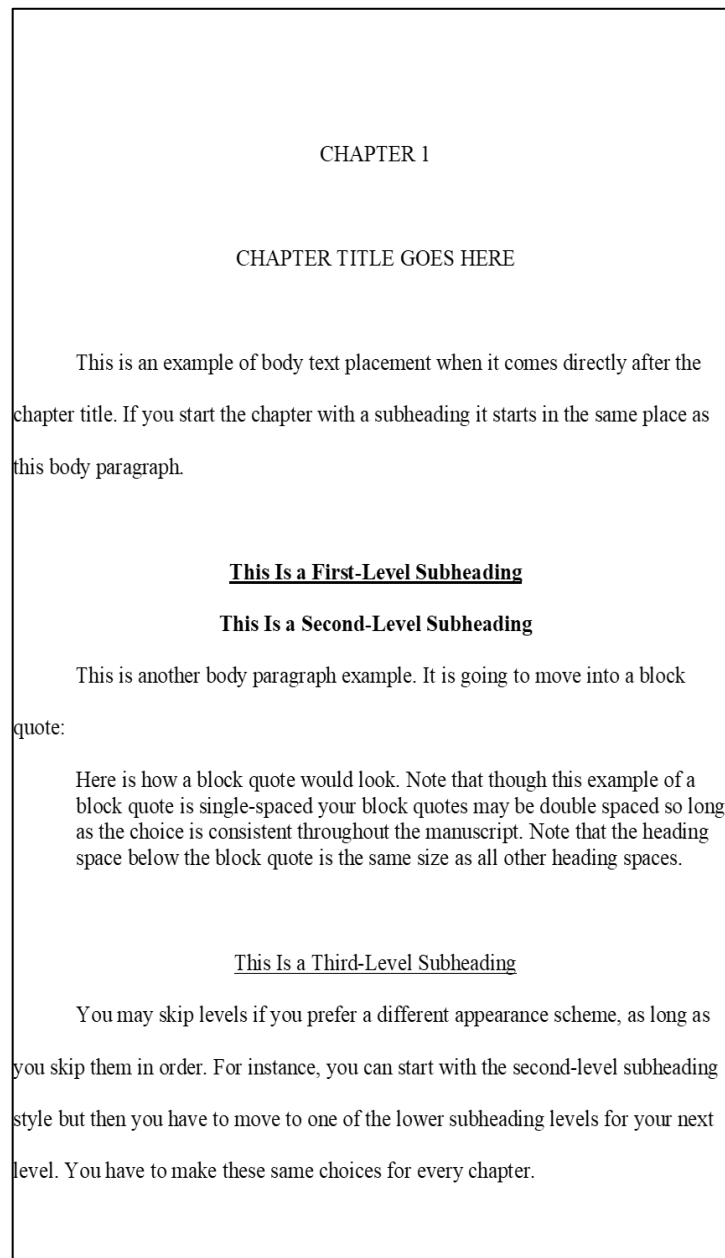
Neither *ser. Androceras* nor *Volucrofolium* is supported as monophyletic in the molecular analyses. These series were paraphyletic in Whalen's (1979a) cladistic analysis and the nonmolecular characters that supported these groups are likely convergent. For instance, *ser. Androceras* was characterized by Whalen (1979a) as having stellate or multangulate ciliate hairs and yellow corollas, lacking flavonoid compounds found in the other two series, and a distribution centered in the central Mexican highlands around Mexico City. These characters are found in species of the Rostratum clade (Fig. 2), but also in *S. ohlsonii*, which does not form a part of this clade. Conversely, Whalen (1979a) placed *S. tribulosum* into *ser. Androceras* despite its pale blue or white corollas. Support and resolution along the backbone of the tree obtained here is weak or lacking, precluding firm conclusions about character evolution in sect. *Androceras* based on the most parsimonious trees. However, constraining all three series each to be monophyletic as well as constraining the taxa of Whalen's *ser. Androceras* and *Volucrofolium* individually to be monophyletic resulted in trees significantly different than the most parsimonious trees from the combined dataset. This further indicates that these two series are likely nonmonophyletic and that the characters that Whalen proposed to diagnose them have evolved multiple times. On the other hand, when all yellow-flowered taxa (i.e. species of *ser. Androceras* minus *S. tribulosum*) were constrained to monophyly, the constrained trees were not significantly different than nonconstrained trees. Therefore, the hypothesis of a single origin of yellow corollas within sect. *Androceras* cannot be rejected.

According to Whalen (1979a), nine of the species of sect. *Androceras* are taprooted annual herbs with wide edaphic tolerances. *Solanum ohlsonii*, *S. leuipipes*, and *S. tribulosum*, however, are calciphilic herbaceous perennials. Judging from their widely separated positions on the molecular trees, it appears that the latter traits evolved independently in the three species.

**Biogeographical Relationships**—Based on his interpretation of cladistic relationships in sect. *Androceras*, Whalen

Figure 2.9 Sample of a page from a previously published article inserted as a chapter in the manuscript.

Note the page number in the top right corner is consecutive with the document and the subheads, tables, figures in previously published articles will be listed in the table of contents, list of figures, list of tables as required.



**Figure 2.10 Sample of the subheading scheme.**

Note the placement of the main headings, the consistent spacing after each main heading and before each subheading.

Also, note after subheads there is only a double space and that when there is no intervening text between subheads they are separated by a double space.

This figure also serves as an example of how to create a part-title page for a figure that takes up an entire page as well as treatment of figures that continue onto a second page.

# APPENDIX A: USING PUBLISHED ARTICLES AS DISSERTATION CHAPTERS

This appendix provides guidelines for those who wish to include published, accepted, or submitted journal articles as chapters in their dissertation. While this option is available, all content must still follow the University of Utah's formatting and submission requirements.

## General structure

Dissertations that incorporate previously published or accepted journal articles must include the following:

- A single abstract at the beginning of the manuscript, covering the entire work
- A general introduction and general conclusion (recommended, not required)
- Each article as a separate chapter, formatted as described below
- A reference section at the end of each chapter, not a single list at the end of the manuscript

The citation style used in each chapter may differ based on the article's journal format. However, formatting of the manuscript (including margins, spacing, and layout) must comply with this handbook.

## Formatting requirements

Even when journal articles are included, the following University formatting rules apply:

- All pages must be numbered consecutively, even if reprints are used
- Running heads (headers at the top of pages) are not permitted
- Tables and figures should be numbered locally by chapter, such as Table 3.1 or Figure 2.2
- Titles and captions must follow the style in this handbook, not the journal's version

If a previously published article is used as-is, you must request permission to reuse it, and the pages must be formatted to match university standards (including page size and margins).

## Copyright and permission

You must obtain formal permission to reuse or adapt any material that has been previously published, even if you're the primary author.

- A copyright permission form must be submitted to the Thesis Office
- A multiple author release form is required when co-authors are involved
- See the university's Permission to Quote form and Figure 1.1 in this handbook for more information

If the original publisher requires written approval or specific wording, this documentation must be included during manuscript submission.

## Property rights and author contracts

Authors typically retain the copyright to their thesis or dissertation unless it has been formally transferred. However, many journals require authors to sign publication agreements that affect how the article may be reused.

You're responsible for checking the terms of your contracts and contacting the journal or publisher for reuse approval.

**Note:** *If permission is not granted, the student may be required to remove or adapt the article prior to graduation.*

## Copyright duration

[Table 1.1](#) (adapted from the U.S. Copyright Office Circular 15A) provides guidance on copyright duration and public domain status.



## APPENDIX B: ACCESSIBILITY

This appendix outlines the accessibility requirements necessary when submitting a thesis or dissertation to the University of Utah. All visual and non-textual content (such as figures, tables, and images) must be accessible to screen readers and visually impaired readers. The guidelines in this appendix will help ensure your work is compliant with current digital accessibility standards.

### Alternate text for figures and tables

Each figure, table, or non-text object must include alternate text (alt text) that conveys the purpose or meaning of the content. Alt text helps visually impaired readers understand non-text content using screen reader software.

#### Figures

Alt text for figures should not simply restate the caption. Instead, provide a brief, accurate description of the visual elements that are essential for understanding the figure's purpose.

- Do not repeat long blocks of text already included in the body
- Focus on visual meaning (e.g., shape, direction, relationships, or trends)

**Example:** If the caption says “Map of the Wasatch Fault system,” the alt text might say “Map of northern Utah highlighting the Wasatch Fault zone running north-south through Salt Lake City and Provo.”



Figure 1. Scene from *Rear Window*. Raymond Burr and Irene Winston, scene still. Dir. Alfred Hitchcock. Paramount Pictures, 1954.

Suitable alternate text: L.B. Jefferies' view of the Thorwald apartment from his window. Lars can be seen from the larger living room window standing while his wife Anna is seen through the bedroom window to the left sitting up in bed. Both windows are open with the curtains and the blinds drawn. The two open windows are upon a brick building with signs of weathering indicative of the New York City setting. A drainpipe divides the space between the two, figuratively and literally placing a line between their respective spaces.

Figure B.1 Example of appropriate alternate text for a figure.

## Tables

Alt text for tables should summarize the table's purpose and structure, not the content of every individual cell.

- State the number of rows and columns
- Summarize the meaning or relationships being presented

**Example:** "This table shows enrollment trends by department for the last five years. Each row shows a department and each column represents one academic year."

Table 1  
*Wood the Woodchucks Chucked in Experiment 1*

| Woodchuck | Wood<br>chucked (in kg) | %    |
|-----------|-------------------------|------|
| 1         | 423.9                   | 94.2 |
| 2         | 373.0                   | 82.9 |
| 4         | 347.0                   | 77.1 |
| 6         | 411.3                   | 91.4 |

*Note.* Each virtual woodchuck received a 450-kg woodpile. Woodchucks 3 and 5 were removed from the analysis because they would not chuck wood.

Suitable alternate text: Table 1 details the amount of wood each of the woodchucks in the experiment chucked, ranging from 347.0 kg to 423.9 kg. It also translates the production to a numerical percentage, ranging from 77.1 to 94.2.

Figure B.2 Example of alternate text for a table.

## Color contrast requirements

Color is often used to communicate data or highlight information, but poor color contrast can make content inaccessible. The following section outlines how to check and improve contrast for all visual elements.

### Guidelines

- Do not rely on color alone to convey meaning (e.g., red vs. green status indicators)
- Use high contrast combinations to ensure legibility
- Ensure that embedded charts and labels have sufficient contrast when printed in grayscale

**Recommended testing methods**

You should use both of the following contrast checking methods:

**Grayscale method**

- Print or convert your figure to grayscale
- Make sure all items remain distinguishable
- Works best for assessing visual contrast in simple graphics

**Contrast analyzer software**

Use online tools such as:

- Colour Contrast Analyser (TPG)
- WAVE (WebAIM)
- Monsido

These tools test foreground/background contrast based on WCAG accessibility standards

Figures B.3 - B.6 illustrate effective and ineffective examples of color contrast in charts and figures.

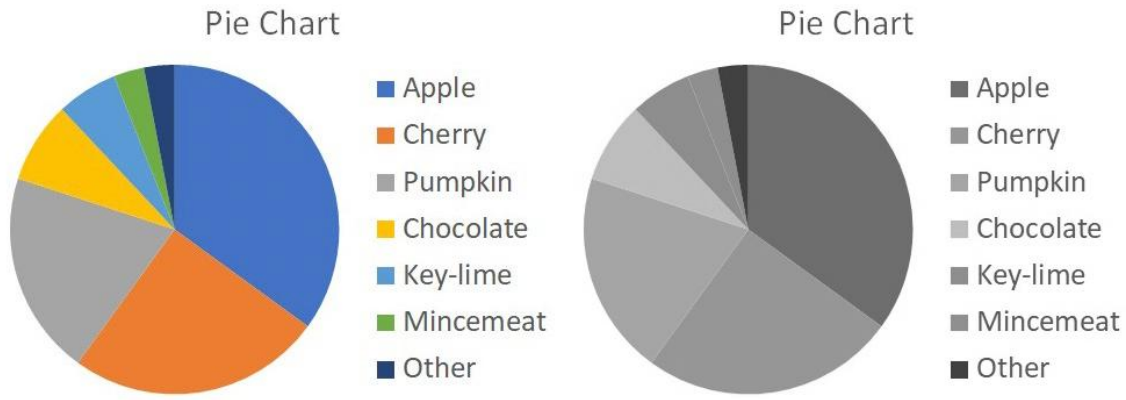


Figure B.3 Unacceptable contrast using the black & white/grayscale method.

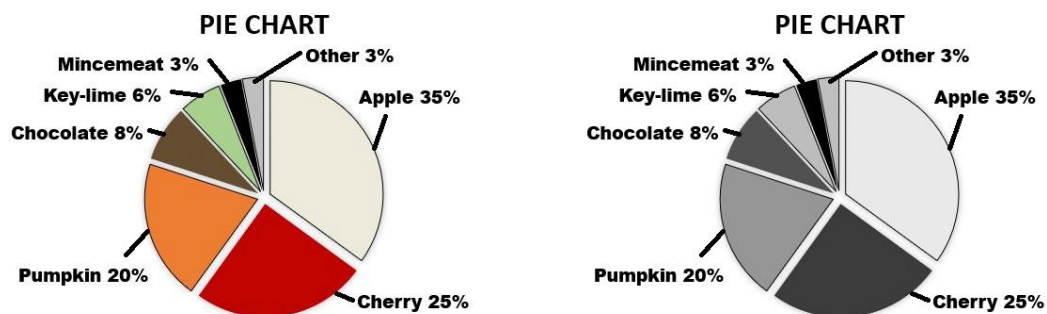


Figure B.4 The same information improved by using colors with more contrast, using borders between segments, and labeling each segment separately.

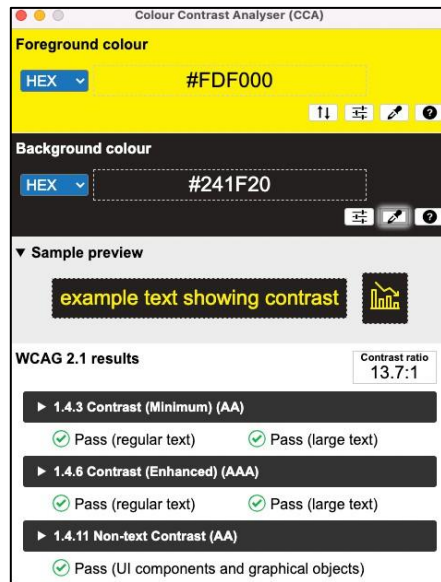


Figure B.5 Acceptable contrast displayed using Colour Contrast Analyser.

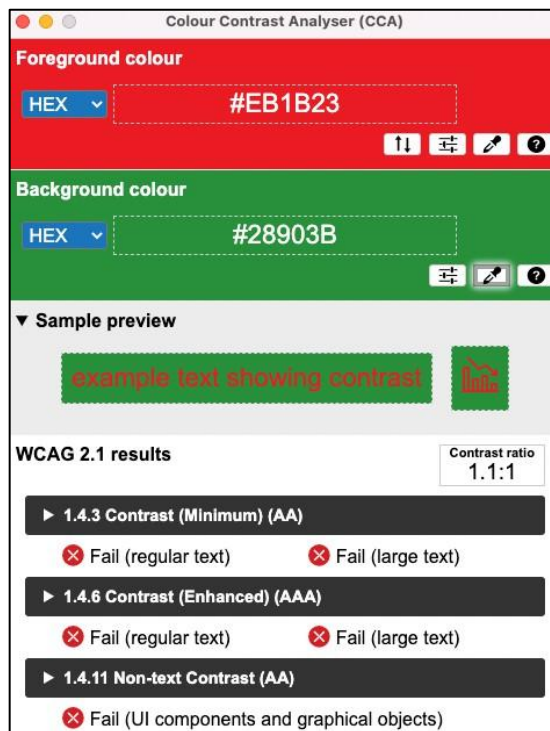


Figure B.6 Unacceptable contrast displayed using Color Contrast Analyser.

## Accessible table formatting

To ensure tables are screen reader–friendly and structurally clean:

- Avoid merged or split cells
- Do not use nested tables
- Remove blank rows and columns that serve no purpose
- Use clear headers for rows and columns
- Keep formatting consistent

These practices ensure that screen readers can correctly interpret the layout and relationships within your table.

## APPENDIX C: COMPARISON OF SPACING REQUIREMENT

| Heading space                                                                                                                                                                                                                                                                                             | Double space                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Single space                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• After main headings</li> <li>• Before all freestanding subheadings</li> <li>• In the case where a figure or table has been placed on a page with body text, a heading space should separate the figure or table from the text on the rest of the page</li> </ul> | <ul style="list-style-type: none"> <li>• Between lines of text</li> <li>• After all freestanding subheadings</li> <li>• Before paragraph subheadings</li> <li>• Before and after single-spaced block quotes</li> <li>• Between lines of subheadings when subheadings extend to more than one line</li> <li>• Between entries in lists of tables and figures and in references</li> <li>• Before and after main heading entries in the table of contents</li> </ul> | <ul style="list-style-type: none"> <li>• Within entries in lists of figures and tables</li> <li>• Within entries in reference lists</li> <li>• Between footnote or endnote entries (double spaces may be used if there are superscript or subscript numbers in the notes)</li> <li>• Between subheads in table of contents</li> <li>• Between lines in figure captions and table titles (double spaces may be used if there are super or subscript numbers in caption or title)</li> </ul> |



## APPENDIX D: ALTERNATE FIGURE AND TABLE PLACEMENT

This appendix outlines acceptable ways to place figures and tables within chapters, especially when embedded near the text that references them. While the standard method is to group figures and tables at the end of each chapter, these alternate placement options may be used if done correctly and consistently.

### Placement on separate pages

The most common method for embedded figures or tables is to place each on its own page with no surrounding text. If multiple small figures or tables are referenced close together in the same paragraph or on the same page, they may be grouped on a single page. In that case:

- There must be no body text on the page
- Each item must be separated by a space larger than a double space
- The page must appear immediately after the one on which the items are first mentioned

### Placement on a page with body text

Tables or figures may appear on a page that also contains body text, but only under the following conditions:

- The table or figure must be referenced first in the body text
- It must either begin or end the page
- There must be at least six consecutive lines of text on that page in addition to the figure or table
- The table or figure must be separated from the body text by a heading space

A table or figure may not be placed mid-page between two blocks of text.

If there is enough space at the bottom of the page where the table or figure is first mentioned, it should appear there. If there is not enough room, it should appear at the top of the following page.

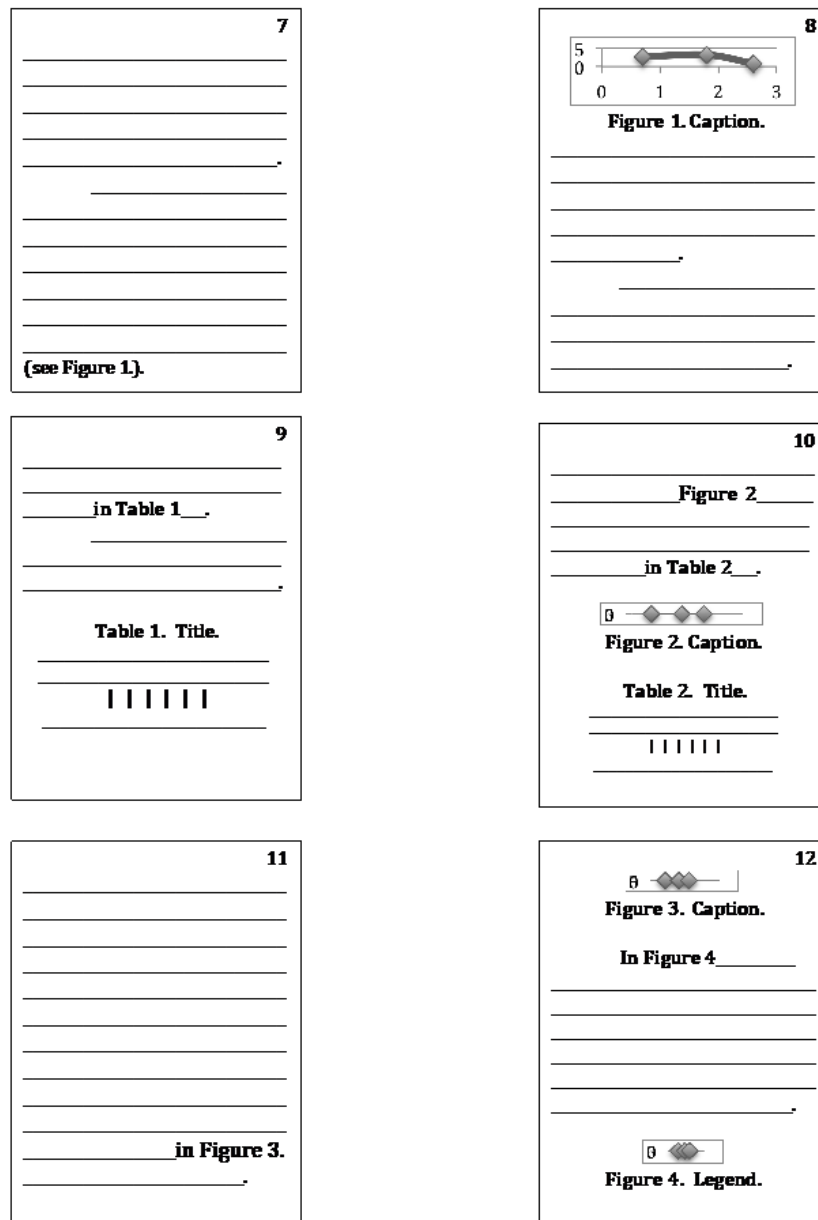
A table or figure that spans more than one page may not be placed on a page that includes body text.

**Tables or figures that continue onto multiple pages**

When a table or figure continues onto a second page:

- The second page must include the number and the word “continued”
  - This appears above the table or below the figure
- For tables, column headings must be repeated on each continued page

***Important note:*** *If these rules are not followed in the first submitted draft, the Thesis Office will require you to place all tables and figures at the ends of their respective chapters or sections.*



**Figure D.1 Illustration of table/figure placement.**

Lines represent lines of text/sentences; note the page numbers in the top right corner. Tables/figures start or end pages as soon as possible after being mentioned (they are not sandwiched between text) and there are at least six of lines on the text with each table/figure.

# Glossary

**Block quotation:** A single quotation that is greater than three lines in length. Block quotes are block indented from the left and either single or double spaced. They are not enclosed in quotation marks.

**Body text:** The narrative or descriptive text that comprises most of your manuscript. This is all text that is not: headings, subheadings, figure captions, table titles, figure or table notes, footnotes, endnotes, equations, algorithms, and so forth.

**Double space:** Extra space between lines of text. The majority of this handbook is double-spaced. In this glossary, there is a double space between each entry.

**Em/En dashes:** Punctuation that can be used in place of commas, parentheses, or colons. In many style guides, an en dash is used to indicate a range (e.g., 10–20) or to give equal weight between two items (e.g., a London–Chicago flight). An em dash is used similarly to a comma or colon to amplify a digression from the text (e.g., Studies—published and unpublished—are included). Consult your style guide for exact uses. An em dash (—) is longer than a hyphen (-) or en dash (–).

**Embargo:** A period of time after your manuscript has been accepted for publication before it is released to the public.

**Final Reading Approval form:** This is one of two forms gathered electronically by the OnBase manuscript submission system when the defended manuscript is submitted to the Thesis Office for format approval. The supervisory committee chair signs to indicate that a final reading has been performed and no more content changes are required. The department/college/school chair/dean signs to indicate that all the appropriate departmental procedures and policies were followed. The signature also indicates that the manuscript is free from grammatical and format errors. The Thesis Office obtains the signature of the Dean of The Graduate School after the editing process is completed.

**Format approval:** Indicates that the thesis or dissertation may be uploaded to ProQuest to begin the thesis release process.

**Format approval process:** Refers to the back-and-forth process in which students 1) submit a document, 2) receive corrections from an editor, 3) make the edits and resubmit a revised document to the editor. This process is repeated until format approval is granted.

**Freestanding subheading:** A subheading that appears on its own line.

**Gaps (see white space):** Spaces between text that have no relation to the end of the chapter, margins, or spacing scheme, for example, an extra space between paragraphs and a large margin or white space at the bottom of the page when it is not the end of a chapter. Some word processing programs may have a default to automatically add a gap between each paragraph. This paragraph setting should be turned off.

**Heading space:** A space that is larger than the double spaces used in a document. Heading spaces are used below main headings and above subheadings. Regardless of the word processing program being used to write the document, the heading spaces must be created consistently throughout the document so that they are always the same size.

**Headline style:** A capitalization scheme in which the first letter of all major words is capitalized. For example: *This Is Written in Headline Style*. Some style guides differ on which words should be capitalized in headline style, check your style guide for details. Subheadings, figure captions, and table titles must consistently use either headline style or sentence style capitalization.

**Hyphen:** Punctuation used between compound terms, compound adjectives, and syllable breaks at the end of a line of text. Please note that words with common prefixes (e.g., non, re, anti, pro, post, pre, micro, multi, etc.) should be treated as one word except when preceding a capitalized word or number (e.g., post-1970, pre-Renaissance) or when the word could be misunderstood (e.g., un-ionized would be confused with unionized). Hyphens are also generally used when the prefix ends and the base word begins with the same vowel (e.g., meta-analysis).

**Local numbering:** Numbers given to subheadings, tables and/or figures, equations, etc. to indicate the chapter and number within that chapter of a given subheading, table, or figure, etc. For example, a table numbered 2.3 would indicate that it is the third table to appear in Chapter 2.

**Main headings:** Headings that indicate a new chapter or main section of the document. Examples include ABSTRACT, CHAPTER 5, SELECTED BIBLIOGRAPHY, LIST OF FIGURES, and chapter titles (or main sections if not using chapters). Main headings always start on a new page, centered 2 inches from the top, and are in all capital letters.

**Main section:** The main divisions of a manuscript. These divisions can also be called chapters. Main section titles are main headings.

**Paragraph subheading:** A subheading level that serves as the first sentence of a new paragraph and does not require a heading space above. The paragraph subheading can be in either sentence style or headline style capitalization.

**Part-title page:** A page that contains only the title of the next element (i.e., section, appendix, chapter, part, or a large figure or table). The text or element that follows begins at the 1" top

margin of the next page.

**Period leaders:** Evenly spaced periods placed closely together to align information in a list. Period leaders are used most commonly in the Table of Contents, List of Figures, and List of Tables.

**Preliminary review:** A cursory review of a document in which the editor addresses only formatting issues such as margins, spacing scheme, placement of tables/figures, gaps/white space, use of main heading, subheadings, etc. Preliminary reviews are performed by the Thesis Office at the student's request prior to a defense and can be performed on a single chapter.

**Reprint:** Previously published material included in a thesis or dissertation as it appears in its published form. Permission from the copyright holder and multiple author releases from coauthors are required to reprint any previously published material.

**Sentence style:** A capitalization scheme in which only the first letter of the first word is capitalized (also proper nouns and acronyms). For example: *This is written in sentence style*. Subheadings, figure captions, and table titles must consistently use either sentence style or headline style capitalization.

**Single space:** Text with no additional space between lines of text. The individual entries in this glossary are single-spaced but are separated from other entries by a double space.

**Spacing scheme:** The size of single spaces, double spaces, and heading spaces within a document. All of these spaces must be of consistent size, that is, all single spaces are the same size, all double spaces are the same size, and all heading spaces are the same size.

**Statement of Approval form:** A form that takes the place of the Supervisory Committee and Final Reading Approval forms in a thesis or dissertation. It is placed after the copyright page and before the abstract. The student's name, the committee members' names, the dates of their approval, the name of the department/college/school chair/dean, and the student's department are typed on the form. This form does not get signed.

**Style guide:** A manual/journal used to determine the writing style of a document. Manuals of style include IEEE, ACM, AIP, Turabian, ACA, and APA among others. Academic journal author guides are also accepted. Students must follow a style guide approved by their department when writing the manuscript. The style guide determines the method of in-text citations, references, and the use of numerals, terms, etc. The Thesis Office must be informed of which style guide was followed. Chapters prepared for or submitted to journals may use different style guides.

**Subheadings:** Headings that divide chapters or main sections into subsections. Unlike main

headings, subheadings are not in all capital letters and can have several levels, including paragraph subheadings.

**Subsection:** Smaller sections within chapters or main sections. Subsections are indicated by subheadings.

**Supervisory Committee Approval form:** This is one of two forms gathered electronically by the OnBase manuscript submission system when the defended manuscript is submitted to the Thesis Office for format approval. The Supervisory Committee Approval form indicates that 1) the committee has read the thesis or dissertation, 2) the content and style of the thesis or dissertation are acceptable, and 3) by majority vote, the manuscript has been found to be satisfactory.

**Thesis release:** Issued in the student's file in Graduate Student Tracking on Campus Information Services when the final upload to ProQuest has been approved.

**White space (see gaps):** Parts of a page that contain no marks, text, tables, figures, etc. White space is usually used to indicate margins and the end of chapters. Large white space should not appear on the top or bottom of the page except a) when the page is used only for figures or tables (i.e., no text appears on the page), b) when a subheading at the bottom of a page has to be moved to the top of the next page because there is not room for at least two lines of continuous text following the subheading, or c) at the end of the chapter. It is incorrect to have a large white space between text and figures or tables.